EN.553.734 Generative AI & Statistical Methods for Clinical Data Science

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June 2026

This course was taught by Dr. Yanxun Xu in the Fall 2026 Semester. The notes are transcribed by Jonathan Ma and may contain errors and omitted content, so any issues are to be directed towards me.

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0.1 What is Clinical Data Science?

Clinical data science combines clinical knowledge, data engineering, statistics, machine learning, and computation to turn heterogeneous health information into evidence that supports decisions. Its purpose is not merely to fit predictive models. The central goal is to connect a clinically meaningful question to appropriate data, a defensible analysis, a quantified statement of uncertainty, and a decision that can be checked by scientists, clinicians, and regulators.

0.1.1 What do we want to answer?

- Does a treatment work, and what is the magnitude of its effect?

- Is the treatment acceptably safe relative to its benefits?
- Which patients benefit most, and which are at elevated risk?
- Which treatment should a particular patient receive?

The first question is causal, the second requires systematic safety assessment, and the latter two introduce treatment-effect heterogeneity, prediction, and individualized decision making. These questions are related but not interchangeable. For example, a model that predicts outcome accurately need not estimate the causal effect of changing treatment.

0.1.2 Sources of clinical evidence

Clinical evidence is assembled from several complementary sources:

- **Clinical trials:** randomized treatment assignments, prespecified endpoints, safety outcomes, and biomarkers;
- **Real-world clinical data:** electronic health records, insurance claims, registries, and laboratory systems;
- **Biomedical measurements:** medical images, genomics and other omics, wearable devices, and sensors;
- **Scientific knowledge:** protocols, regulatory guidance, publications, clinical guidelines, and prior studies.

These sources differ in structure, provenance, measurement error, missingness, and susceptibility to bias. Randomization can support causal identification in a trial, whereas observational data generally require additional assumptions and adjustment. Unstructured documents may encode essential definitions that are absent from an analysis table.

The evidence pipeline can be summarized as

$$\text{clinical question} \longrightarrow \text{data} \longrightarrow \text{statistics and AI} \longrightarrow \text{evidence} \longrightarrow \text{decision}.$$

Each arrow represents work: defining a target quantity, collecting appropriate variables, constructing analysis-ready data, choosing an estimand and method, validating outputs, and communicating uncertainty.

0.1.3 Biopharmaceutical statistics and drug development

Biopharmaceutical statistics links trial design with statistical analysis. Before observing the trial outcomes, statisticians help determine the target population, treatment arms, endpoints, randomization and stratification scheme, sample size, interim analyses, and error-control strategy. After data are collected, they estimate treatment effects, quantify uncertainty, analyze efficacy and safety, assess robustness, and explain what conclusions the design supports. Questions such as “How many patients are needed?” and “Can external evidence be borrowed?” are therefore design questions as much as analysis questions.

Drug development is a long evidence-generation process:

$$\text{Discovery} \to \text{Preclinical} \to \text{Phase I} \to \text{Phase II} \to \text{Phase III} \longrightarrow \text{Regulatory review} \to \text{Post-market}.$$

Preclinical studies occur before exposure in humans. Phase I emphasizes initial safety, tolerability, pharmacokinetics, and dose; Phase II investigates preliminary efficacy and dose selection; Phase III provides confirmatory benefit–risk evidence in larger populations. Regulatory review evaluates the totality of evidence, and post-market work studies effectiveness and uncommon or long-latency safety outcomes under real-world use.

0.1.4 The work of statisticians and clinical data scientists

The work spans three connected layers:

1. **Design:** understand the disease and protocol, define the population and endpoints, specify estimands, and design the study;
2. **Data:** determine what must be collected, define derivations, reconcile sources, and construct traceable analysis datasets;
3. **Evidence:** analyze efficacy and safety, incorporate prior or external information when justified, perform sensitivity analyses, and communicate the result.

Consequently, the job is not simply to receive one rectangular dataset and run a model. An analysis dataset is interpreted alongside the protocol, statistical analysis plan (SAP), regulatory guidance, clinical definitions, publications, prior trials, and the tables and figures that must be produced.

0.1.5 From statistical software to generative AI

Traditional statistical programs in R, SAS, or similar systems apply explicit human-written rules to structured inputs and produce reproducible structured outputs. Conventional machine learning learns a mapping from data for a particular prediction task. A foundation model is broader: it can accept documents, images, tables, data, code, and instructions and generate language, code, or structured representations for many tasks.

This makes generative AI relevant to workflows such as reading protocols, extracting definitions, interpreting tables, writing analysis code, answering questions across documents, and reconciling multiple information sources. The useful object is not necessarily a stand-alone chatbot. It is often a *system* in which a model calls search, database, calculation, and code execution tools while maintaining a traceable workflow.

Remark 0.1.1 (Prediction, generation, and action)**.** A task-specific predictive model typically maps a fixed input schema to one output. A generative model can produce flexible content from heterogeneous context. An agentic system goes one step further by selecting and invoking tools, observing their outputs, and iterating toward a goal. These additional capabilities also introduce additional failure modes.

0.1.6 Why scientific AI needs more than a plausible answer

In clinical and regulatory work, fluent prose is not sufficient. A reliable workflow should provide:

- **Accuracy:** claims and extracted values agree with authoritative sources;
- **Structure:** important outputs are represented in a defined, machine-readable schema rather than only in prose;
- **Reproducibility:** inputs, model and tool versions, prompts, parameters, and intermediate results are recorded;
- **Validation:** a qualified person or validated procedure can verify the result against source evidence;
- **Tool grounding:** the model retrieves actual documents and executes exact computations instead of relying on memory or approximate verbal arithmetic;
- **Traceability:** every important conclusion can be linked back to data, code, and document locations.

These requirements motivate combining generative AI with tools, statistics, domain knowledge, and explicit quality control. The objective is not to replace statistical reasoning, but to build systems that can assist with complex information work without obscuring assumptions or evidence.

0.1.7 Clinical Protocols

The clinical trial protocol is the document that prespecifies how a trial will be conducted before enrollment begins. A protocol can span roughly 100–300 pages of prose, tables, flow charts, and appendices. It states the scientific objectives, population, treatments, endpoints, visit schedule, procedures, safety rules, and statistical design. It must align clinical science, operations, data collection, and analysis.

The protocol is controlled rather than casually "living." Changes require documented amendments, version control, review, and, where applicable, regulatory and ethics approval. Downstream artifacts—case-report forms, analysis datasets, the SAP, tables, listings, figures, clinical study reports, and submissions—must remain traceable to the governing version.

Conceptually, the protocol connects

$$\text{scientific objective} \leftrightarrow \text{study population} \leftrightarrow \text{treatment} \leftrightarrow \text{endpoints}$$
$$\text{endpoints} \leftrightarrow \text{data collection} \leftrightarrow \text{statistical analysis}.$$

It is therefore a bridge from natural-language clinical decisions to data definitions and executable statistical rules. Protocol extraction is not an end in itself; it is a useful test case because errors at this junction propagate through the entire evidence-generation process.

0.1.7.1 A protocol makes one controlled experiment

A multicenter trial may involve hundreds of patients, investigators, and sites. The protocol makes their contributions part of one interpretable experiment by serving three functions:

- **Standardization:** sites use the same eligibility rules, treatment schedules, assessment procedures, visit timing, and definitions;
- **Control:** key conditions are prespecified, treatment groups are made comparable, and avoidable variation is reduced;
- **Interpretation:** the resulting evidence identifies who was studied, what was done, what was measured, and how comparisons were made.

Without a common protocol, a nominally multicenter trial could become a collection of different experiments. The protocol therefore turns a clinical question into a reproducible data-generating process.

It also combines multiple representations. Narrative sections explain rationale and clinical judgment; eligibility lists express logical rules; tables encode schedules and thresholds; diagrams describe treatment flow; and statistical sections define populations, endpoints, models, and decisions. Correct interpretation often requires reconciling all of them.

0.1.7.2 A functional map of a protocol

Section numbering differs across sponsors, so it is more useful to ask what question each part answers:

Question	Typical content	Purpose
Why?	Background and rationale	Disease context, unmet need, prior evidence, and justification for the study
What?	Objectives, endpoints, hypotheses	Scientific intent, measured outcome, and statistical claim
How?	Design and intervention	Randomization, control, masking, treatment regimens, and stopping rules
Who?	Population and eligibility	The population created by inclusion and exclusion criteria
When?	Assessments and schedule of activities	What is measured at each visit and during follow-up
Analysis?	Statistical design and SAP	Analysis populations, derivations, models, missingness, multiplicity, and decision rules

Public protocol packages may redact proprietary scientific material, personal information, or nonessential appendices. Absence from a public copy does not mean that the information was absent from trial planning. An AI extraction system should distinguish an explicit statement of absence from content that is redacted, omitted, or unavailable, and it should retain the provenance and version of the document it processed.

0.1.7.3 Objective, endpoint, and hypothesis

These terms describe different layers of the same question:

- an **objective** states the scientific question, such as whether treatment improves progression-free survival;
- an **endpoint** defines the outcome variable used to represent that question, including its time origin, event, assessment procedure, and observation window;
- a **hypothesis** states the statistical claim to be tested, such as superiority of one treatment over another for the endpoint.

A strong protocol aligns scientific intent, measurement, and statistical decision. Extracting only an endpoint name such as "PFS" is insufficient: assessment method, reviewer, visit schedule, censoring, and intercurrent events can change the estimand and the interpretation.

Comparator choice is part of the scientific question. An active comparator asks whether a new therapy improves upon an existing treatment, whereas a placebo comparison asks a different question. The appropriate control depends on the treatment landscape at the time of design, eligibility, background therapy, ethics, and the intended regulatory claim.

0.1.7.4 Design and intervention as structured rules

Randomization specifies who receives each treatment and supports causal comparison by balancing baseline factors in expectation. Stratified randomization promotes balance within prespecified prognostic categories and is often mirrored in the primary analysis. Masking reduces behavioral and assessment biases; in an open-label trial, blinded independent outcome review may protect key endpoints even when treatment cannot be concealed.

A treatment arm is not merely a drug name. It is a structured regimen that includes dose, route, frequency, treatment duration, allowable timing windows, criteria for holding or discontinuing treatment, retreatment, concomitant and prohibited medications, and toxicity-management rules. These elements must be represented if protocol text is converted into executable logic.

0.1.7.5 Eligibility is executable clinical logic

Eligibility criteria protect participants, define the population to which the trial directly applies, and reduce factors that would obscure treatment effects or make assessments unreliable. Applying them to real candidates produces the enrolled population, which may be narrower than the target clinical population. This distinction later matters for generalizability, transportability, external controls, and evidence synthesis.

Criteria can include nested conditions, temporal windows, laboratory thresholds, units, local upper limits of normal, and exceptions. A useful machine-readable representation separates:

variable, operator, threshold, unit, time window, exception, source citation.

For example, “AST/ALT at most 2.5 times ULN, or at most 5 times ULN with liver metastases” is a conditional rule, not one numeric cutoff. Negation and scope also matter: “no treatment within 14 days” differs from “no prior treatment.”

0.1.7.6 The schedule of activities specifies future data

The schedule of activities is a visit-by-assessment matrix. Rows identify procedures such as imaging, laboratory tests, adverse-event collection, pharmacokinetics, and patient-reported outcomes; columns identify screening, treatment visits, end of treatment, and follow-up. Each cell determines whether and when a datum should exist.

Thus the schedule propagates into the eventual analysis:

$$\text{study-flow requirement} \longrightarrow \text{clinical record or CRF} \longrightarrow \text{analysis variable} \longrightarrow \text{endpoint derivation} \longrightarrow \text{statistical result.}$$

For a time-to-event endpoint, imaging dates, lesion assessments, progression status, death dates, missed visits, and central review may determine the event indicator and censoring time. Timing can follow the study calendar rather than shift automatically when a treatment cycle is delayed.

0.1.7.7 Efficacy, safety, and patient experience

Efficacy measurements require both a clinical assessment framework and an endpoint interpretation. Tumor response, for example, depends on imaging modality, baseline lesions, assessment frequency, response criteria, and who performs the review. Progression-free survival may combine those assessments with death information, whereas overall survival is derived from death dates.

Safety is a repeated monitoring system rather than one endpoint. It combines adverse events and severity grades, laboratory tests, vital signs, examinations, electrocardiograms, and events of special interest. Protocol rules may map a toxicity grade and attribution to an action such as holding treatment, restarting after recovery, or permanent discontinuation. Such rules are natural candidates for structured decision logic, but ambiguity must be escalated rather than silently resolved by a model.

Patient-reported outcomes and health-economic measures may impose additional ordering and timing requirements. For example, a quality-of-life questionnaire may need to be completed before drug administration, adverse-event discussion, or disclosure of disease status to reduce measurement bias.

0.1.7.8 Analysis populations and statistical commitments

An endpoint result is always about an analysis population. The intention-to-treat population generally includes all randomized participants according to assigned treatment and preserves the randomized comparison. A safety population may instead analyze all treated participants according to treatment received. These choices are part of what a result means, not merely data-management labels.

Sample size and power encode design assumptions about the control outcome, clinically meaningful effect, variability or event rate, accrual, follow-up, dropout, Type I error, and desired power. A survival trial can be event-driven: enrollment n matters, but the primary analysis occurs after a prespecified number of events or amount of follow-up. Interim analyses and multiplicity rules additionally define when evidence is examined and how error is controlled across endpoints, treatment arms, and repeated looks. In this sense, part of a protocol is a prespecified decision algorithm for accumulating evidence.

0.1.7.9 Cross-document traceability and AI challenges

The protocol belongs to a larger chain:

$$\text{protocol} \to \text{SAP} \to \text{CRF/raw data} \to \text{analysis data}$$
$$\to \text{code and output} \to \text{clinical study report and regulatory evidence.}$$

The same concept may appear in many places. A PFS objective may connect to the imaging schedule, response criteria, event and censoring definitions in the SAP, derived analysis variables, an interim boundary, and the final Kaplan– Meier curve, hazard ratio, confidence interval, and test. Correct extraction therefore requires cross-section and eventually cross-document reasoning.

Protocols are difficult for computers because they are long, use specialized and sometimes ambiguous language, distribute meaning across prose and tables, contain cross-references, and encode temporal logic and exceptions. A robust AI workflow should preserve document coordinates, resolve references, normalize units and terminology, represent uncertainty, validate structured outputs against a schema, and route consequential or conflicting statements for human review.

0.1.8 An Example Clinical Protocol

The example is protocol MK-3475-006, a multicenter, randomized, controlled, open-label, three-arm Phase III trial in advanced melanoma. It compares twodosing schedules of MK-3475 (pembrolizumab) with ipilimumab. Approximately 645 patients are randomized in a 1:1:1 ratio, with stratification by line of therapy, PD-L1 expression, and ECOG performance status. The co-primary endpoints are progression-free survival (PFS) and overall survival (OS); the principal secondary efficacy endpoint is overall response rate (ORR).

How to read a protocol. A protocol should be read as a chain from the clinical question to the final statistical claim, not simply from the first page to the last:

1. **Check identity and history.** Record the protocol number, amendment number and date, and distinguish the original protocol from the final version. Read the summary of changes first. In this example, an amendment changed the primary ORR population to intention-to-treat (ITT) following FDA feedback. Such a change can alter the interpretation of the results and must not be silently mixed with an earlier analysis plan.

2. **Translate the clinical question into estimands.** Identify the population, treatments, endpoints, time origin, intercurrent events, and treatment contrast. Here, PFS is time from randomization to centrally assessed progression or death, while OS is time from randomization to death from any cause. Similar endpoint names do not guarantee identical definitions across studies.

3. **Connect objectives, hypotheses, and endpoints.** Separate primary, secondary, and exploratory objectives. The trial succeeds if at least one MK-3475 arm is superior to ipilimumab for PFS at an interim analysis or for OS at an interim or final analysis. This logical rule is more precise than the informal statement that the treatment "works."

4. **Understand who enters the trial and how treatment is assigned.** Review inclusion and exclusion criteria, recruitment setting, randomization ratio, stratification factors, masking, treatment duration, crossover or subsequent therapy, and discontinuation rules. These details determine the target population, internal validity, generalizability, and possible sources of bias. The open-label design makes the blinded independent central review of PFS and ORR especially important.

5. **Follow the schedule of assessments.** The flow chart determines when scans, laboratory measurements, adverse-event assessments, and follow-up occur. Visit windows and missed assessments affect the observed event time, missingness, and censoring; they are therefore part of the statistical design rather than merely operational details.

6. **Audit the statistical analysis plan (SAP).** Match each endpoint to its analysis population, statistical method, missing-data rule, and sensitivity analyses. In this trial, efficacy uses ITT; safety uses the all-patients-as-treated population. PFS and OS use stratified log-rank tests and stratified Cox models, while ORR uses the stratified Miettinen–Nurminen method. Patients with missing ORR are treated as nonresponders.

7. **Inspect censoring, multiplicity, and interim monitoring.** For primary PFS analysis, a patient without progression or death is censored at the last disease assessment. Alternative rules address missed scans, new anticancer therapy, and investigator assessment. The overall one-sided Type I error is controlled at 2.5% across two experimental arms, PFS and OS, and repeated looks using alpha allocation, Bonferroni adjustment, sequential testing, and a Hochberg procedure. Looking only at nominal p-values would ignore this testing structure.

8. **Reconstruct the sample-size argument.** The study is event-driven, with about 435 deaths planned for the final OS analysis. Check the assumed control survival, clinically meaningful hazard ratio, accrual, follow-up, dropout, power, and early-stopping rules. These assumptions should later be compared with the realized trial data.

9. **Read safety as carefully as efficacy.** Confirm adverse-event definitions, severity grading, reporting windows, exposure adjustment, and prespecified events of special interest. The protocol uses a tiered safety analysis; not every confidence interval or nominal p-value is a formal multiplicity-controlled hypothesis test.

A useful practical check is to build a traceability table with one row per objective or endpoint and columns for the endpoint definition, analysis population, treatment contrast, observation window, intercurrent-event and missing-data handling, statistical method, multiplicity allocation, sensitivity analyses, and protocol/SAP citation. Any blank or contradictory cell becomes a question for the statistician and clinician rather than an invitation for the model to guess.

0.2 Early-Phase Clinical Trial Design

Early-phase trials decide which drug candidates and doses deserve further development. Patients and time are scarce, outcomes are noisy, and each new observation may change what is ethically appropriate for the next participant. The design must therefore connect a clinical question to an estimand, an allocation rule, stopping decisions, and prespecified operating characteristics.

The phases have different emphases:

- **Phase I** studies initial safety, tolerability, pharmacokinetics (PK), pharmacodynamics (PD), and dose. In oncology it often seeks a maximum tolerated dose (MTD) or a recommended Phase II dose.
- **Phase II** screens for preliminary efficacy while refining safety and dose. A single-arm Phase IIa study may compare a response rate with a historical benchmark; a randomized or multi-arm Phase IIb study may compare treatments or doses.
- **Phase III** supplies confirmatory benefit–risk evidence, usually through a larger randomized controlled design.
- **Phase IV** studies effectiveness, rare or delayed harms, and additional uses after marketing.

These descriptions are typical rather than universal. The design should be understood from its objectives and estimands, not from a phase label alone.

0.2.1 Bayesian updating and operating characteristics

For parameter θ, prior density $\pi(\theta)$, and data D with likelihood $L(D \mid \theta)$, Bayes' rule gives

$$\pi(\theta \mid D) = \frac{L(D \mid \theta)\pi(\theta)}{\int L(D \mid u)\pi(u)\ \mathrm{d}u} \propto L(D \mid \theta)\pi(\theta).$$

The posterior can answer trial-specific questions such as $\mathbb{P}(\theta > \theta_0 \mid D)$ and supplies the posterior predictive distribution of future outcomes,

$$p(\widetilde{D} \mid D) = \int p(\widetilde{D} \mid \theta)\pi(\theta \mid D)\ \mathrm{d}\theta.$$

Bayesian decision rules should still be evaluated over repeated simulated trials. Important frequentist operating characteristics include Type I error, power, probability of selecting

each dose, expected sample size, early-stopping probability, patient allocation, and toxicity exposure. Thus Bayesian updating and frequentist design evaluation are complementary rather than mutually exclusive.

External evidence can enter through an informative prior, which is valuable when early-phase samples are small. Borrowing is credible only to the extent that populations, measurements, regimens, and study conduct are commensurate. Prior-data conflict can bias decisions, so protocols should prespecify robust or discounted borrowing and report sensitivity to the prior.

0.2.2 Phase I dose finding

Suppose $d_1 < \cdots < d_J$ are ordered dose levels and $Y_i = 1$ indicates a protocol-defined dose-limiting toxicity (DLT) during a fixed assessment window. Let

$$p_j = \mathbb{P}(Y = 1 \mid d_j)$$

and let ϕ be the target toxicity probability. A common statistical target is the dose whose toxicity probability is closest to ϕ,

$$j^* = \arg\min_{1 \le j \le J} |p_j - \phi| ,$$

subject to overdose-control and clinical constraints. Calling this dose the MTD is a model-based convention; the eventual recommended dose may also depend on PK, PD, delayed toxicities, efficacy, formulation, and feasibility.

The monotonicity assumption $p_1 \le \cdots \le p_J$ is clinically plausible for many cytotoxic agents but is still an assumption. The DLT definition, grading, attribution, observation window, evaluability, and handling of incomplete follow-up must be specified before the allocation algorithm is meaningful.

0.2.2.1 The rule-based 3+3 design

The traditional 3+3 design treats cohorts at one dose at a time:

- 0/3 DLTs: escalate one dose level;
- 1/3 DLTs: expand the current dose to six participants;
- 1/6 DLTs: typically escalate;
- at least 2/3 or 2/6 DLTs: stop escalation and de-escalate.

Under a conventional implementation, the estimated MTD is generally the highest previously tested lower dose with no more than one DLT among six, not the dose at which excessive toxicity was observed. Boundary cases and the number required at the lower dose must be explicitly defined by the protocol.

The design is transparent and easy to implement, but its dose recommendation is based on very small local counts. It does not use a formal target *phi*, borrow information smoothly across doses, quantify uncertainty, or optimize a well-defined statistical criterion. Escalation can be slow, and different true dose–toxicity curves can lead to poor selection probabilities.

0.2.2.2 Continual reassessment method

The continual reassessment method (CRM) links all doses through a parsimonious model. Let $0 < s_1 < \cdots < s_J < 1$ be a prespecified toxicity *skeleton*. A common one-parameter power model is

$$p_j(a) = s_j^{\exp(a)}. \tag{0.2.1}$$

At $a = 0$, $p_j(a) = s_j$. Because $0 < s_j < 1$, increasing a increases the exponent and therefore *decreases* $p_j(a)$; decreasing a increases the fitted toxicity probabilities. This direction is easy to misstate, although alternative parameterizations may reverse it.

If patient i receives dose index j_i and has observed DLT y_i, the likelihood after n evaluable participants is

$$L_n(a) = \prod_{i=1}^{n} p_{j_i}(a)^{y_i}\{1 - p_{j_i}(a)\}^{1-y_i}.$$

With prior $\pi(a)$, the posterior is proportional to $L_n(a)\pi(a)$. One CRM allocation rule estimates

$$\widehat{p}_{j,n} = \mathbb{E}[p_j(a) \mid D_n]$$

and assigns the next cohort to the admissible dose minimizing $|\widehat{p}_{j,n} - \phi|$. The final recommendation can use the same rule after all outcomes have been observed.

The one-parameter model is intentional: a Phase I sample is too small to estimate an unconstrained probability at every dose. Linking doses lets an outcome at one level update beliefs about all levels. The cost is model dependence, making skeleton choice, prior calibration, and scenario simulation essential.

0.2.2.3 Safety modifications

A model recommendation is not automatically an ethically permissible action. Common restrictions include:

- begin at a clinically justified low dose;
- prohibit skipping untried dose levels during escalation;
- prohibit escalation immediately after an observed DLT;
- require cohorts rather than individual-patient escalation;
- stop or exclude a dose if its posterior overdose probability exceeds a threshold, for example $\mathbb{P}(p_j > \phi + \epsilon \mid D_n) > \gamma$;
- pause when DLT outcomes are pending or when non-DLT safety information requires review.

These restrictions may reduce statistical efficiency but improve protection against rapid escalation and delayed information. They are part of the design and must be included in simulation; evaluating an unconstrained CRM does not validate the constrained trial that will actually run.

0.2.2.4 Simulation of operating characteristics

For each clinically plausible true curve $p^{(m)} = (p_1^{(m)}, \ldots, p_J^{(m)})$, simulation proceeds as follows:

1. initialize the design exactly as specified in the protocol;
2. assign a dose using the current rule;
3. simulate $Y_i \sim \text{Bernoulli}(p_{j_i}^{(m)})$ and any outcome delays;
4. update the posterior and apply safety and stopping rules;
5. continue until the trial terminates and record the recommendation;
6. repeat many independent trials.

The resulting summary should include selection probability for every dose, probability of selecting above the target, mean allocation by dose, mean DLT count and rate, early stopping, and sample size. Correct-selection probability alone can hide an unsafe design. Conversely, a design that treats more patients near the true MTD may have a higher observed DLT rate than an overly conservative design, so toxicity exposure and informativeness must be interpreted jointly.

0.2.3 Phase II screening designs

For a binary response endpoint with probability p, a single-arm Phase II design often separates an uninteresting rate p_0 from a promising rate p_1:

$$H_0 : p \le p_0, \qquad H_1 : p \ge p_1, \qquad p_1 > p_0.$$

The design aims to bound Type I error by *alpha* at p_0 and achieve power at least $1 - \beta$ at p_1. Because a single-arm comparison relies on a historical benchmark, changes in eligibility, assessment, supportive care, and case mix can invalidate the interpretation even if the binomial calculations are correct.

0.2.3.1 Simon two-stage design

Let $X_1 \sim \text{Binomial}(n_1, p)$ be the number of Stage 1 responses. Stop for futility if $X_1 \le r_1$; otherwise accrue to total sample size N and declare the treatment promising if the total response count $X = X_1 + X_2$ is at least r, where $X_2 \sim \text{Binomial}(N - n_1, p)$ independently. The probability of a positive decision at response probability p is

$$\Psi(p) = \sum_{x=r_1+1}^{n_1} \binom{n_1}{x} p^x (1-p)^{n_1-x} \mathbb{P}\{\text{Binomial}(N - n_1, p) \ge r - x\}. \tag{0.2.2}$$

Thus the exact design constraints are

$$\Psi(p_0) \le \alpha, \qquad \Psi(p_1) \ge 1 - \beta.$$

The expected sample size at response probability p is

$$\mathbb{E}_p[N_{\text{trial}}] = n_1 + (N - n_1)\mathbb{P}_p(X_1 > r_1). \tag{0.2.3}$$

Candidate integer quadruples (n_1, N, r_1, r) can be enumerated. A Simon *optimal* design minimizes (0.2.3) at p_0 subject to the error constraints; a *minimax* design minimizes the maximum sample size N, usually with expected sample size used to break ties. This makes the tradeoff explicit: early futility can save participants when the treatment is inactive, while the maximum sample size preserves power when it is promising.

0.2.4 Bayesian predictive probability of success

Bayesian monitoring can ask whether continuing enrollment is likely to produce a successful final analysis. Consider a beta-binomial model

$$p \sim \text{Beta}(a, b), \qquad X \mid p \sim \text{Binomial}(n, p).$$

After x responses among n participants,

$$p \mid x, n \sim \text{Beta}(a + x, b + n - x).$$

If $m = N - n$ participants remain and Y is their future response count, then integrating over the posterior gives the beta-binomial predictive mass

$$\mathbb{P}(Y = y \mid x, n) = \binom{m}{y} \frac{B(a + x + y, b + N - x - y)}{B(a + x, b + n - x)}, \qquad y = 0, \ldots, m. \tag{0.2.4}$$

Suppose final success is defined by a posterior rule

$$\mathbb{P}(p > p_0 \mid X_N, N) > \eta.$$

Let $S(z)$ be the indicator that this rule holds when the final total is z. The predictive probability of success (PPS) at the interim is then

$$\text{PPS}(x, n) = \sum_{y=0}^{m} S(x + y)\mathbb{P}(Y = y \mid x, n). \tag{0.2.5}$$

This calculation averages over both future binomial variation and current uncertainty about p. A design might stop for futility when PPS is below a lower boundary, continue in an intermediate region, and possibly stop for success above an upper boundary.

Continuous monitoring is not permission to improvise after seeing the data. The prior, success rule, monitoring frequency, maturity requirements, boundaries, and handling of delayed or missing outcomes must be prespecified. The complete algorithm should then be simulated under p_0, p_1, and intermediate or misspecified scenarios to estimate Type I error, power, expected sample size, and stopping behavior.

0.2.5 Multi-arm Phase IIb designs

Phase IIb studies may compare several doses or regimens while continuing to learn which arm should advance. Equal randomization protects balance and gives stable comparisons. Outcome-adaptive randomization (OAR) instead changes future allocation probabilities as outcomes accumulate, seeking to learn while assigning more participants to promising arms.

Let p_k be the response probability for active arm k and let D_t denote data available at adaptation time t. Define the posterior best-arm probability

$$q_{k,t} = \mathbb{P}\Big(p_k = \max_{\ell \in \mathcal{A}_t} p_\ell \,\Big|\, D_t\Big),$$

where $\mathcal{A}_t$ is the set of active arms. A tempered OAR rule is

$$\pi_{k,t} = \frac{q_{k,t}^{c_t}}{\sum_{\ell \in \mathcal{A}_t} q_{\ell,t}^{c_t}}. \tag{0.2.6}$$

When $c_t = 0$, allocation is equal; $c_t = 1$ allocates in proportion to posterior best-arm probability; and larger values concentrate allocation more strongly. Because early $q_{k,t}$ values are noisy, practical designs use an equal- allocation burn-in, a minimum probability floor, and gradual tempering. With floor $\epsilon < 1/K_t$ for $K_t = |\mathcal{A}_t|$, one implementation is

$$\widetilde{\pi}_{k,t} = \epsilon + (1 - K_t\epsilon)\pi_{k,t},$$

which preserves $\sum_k \widetilde{\pi}_{k,t} = 1$ and guarantees every active arm at least ϵ allocation.

At each cohort the design makes two distinct decisions: whether each arm remains active and how the next participants are allocated among active arms. For example, an arm may be suspended for futility if

$$\mathbb{P}(p_k > p_{\min} \mid D_t) < \gamma_F,$$

or declared sufficiently promising if a prespecified superiority or target- exceedance probability crosses an upper boundary. These rules, including what happens when all arms stop or several appear promising, must be simulated as one complete algorithm.

OAR does not automatically improve power or ethics. Rapid adaptation can lock onto an arm favored by chance, reduce concurrent information on other arms, create time trends in treatment comparisons, and behave poorly with delayed outcomes. Evaluation should report allocation distributions and participant outcomes in addition to arm-selection probability, power, Type I error, and sample size.

0.2.5.1 Dose response and the optimal biological dose

The most efficacious dose need not be the highest dose. For targeted and biologic therapies, benefit can plateau while toxicity or burden continues to increase. Let $m(d;\theta)$ be a flexible dose–response curve with placebo or baseline response $m(0;\theta)$ and modeled maximum $m_{\max}(\theta)$. A common efficacy summary is

$$\mathrm{ED}_{95}(\theta) = \inf\Big\{d : m(d;\theta) - m(0;\theta) \geq 0.95\Big[m_{\max}(\theta) - m(0;\theta)\Big]\Big\}.$$

This is not automatically the optimal biological dose (OBD). An OBD rule may combine posterior efficacy, toxicity, PK/PD, durability, and feasibility, for example selecting the lowest admissible dose near the efficacy plateau. A flexible response model can borrow information across adjacent doses, but its shape assumptions and the loss attached to incorrect selection require scenario-based sensitivity analysis.

0.2.5.2 Delayed outcomes

When the endpoint is slow, naive OAR updates only on participants with complete follow-up. If recruitment is fast, the effective information can lag far behind enrollment and allocation may be driven by a small, unrepresentative subset. One remedy is a joint model for an early outcome R_i and later time-to-event outcome (T_i, Δ_i):

$$p(R_i, T_i, \Delta_i \mid Z_i = k, \vartheta) = p(R_i \mid Z_i = k, \vartheta)p(T_i, \Delta_i \mid R_i, Z_i = k, \vartheta).$$

Participants with censored PFS still contribute survival likelihood through $S_k(C_i)$, while the association between early response and PFS determines how much the early signal informs later benefit. Misspecifying that association can misdirect allocation, so designs should compare joint-model, mature-only, and robust fallback behavior under differing delays and correlations.

0.2.6 Decision-theoretic monitoring

A decision-theoretic design specifies a state, available actions, uncertain consequences, and utility. At time t, let actions include continuing, abandoning the treatment, or stopping and recommending it. Given posterior $\pi(\theta \mid D_t)$, the Bayes action maximizes posterior expected utility:

$$a_t^* = \arg\max_{a \in \mathcal{A}_t} \mathbb{E}\{U(a, \theta, \text{future data}) \mid D_t\}. \tag{0.2.7}$$

Utility can value outcomes for current trial participants and future patients while penalizing recruitment, delay, toxicity, and incorrect recommendations. A weight λ may control the relative value assigned to current versus future patients, but choosing it is a scientific and ethical judgment rather than a purely computational step.

Full dynamic programming is often infeasible because the value of continuing depends on all future adaptations. A practical approximation parameterizes decision boundaries using posterior summaries. If m_t and s_t are the posterior mean and standard deviation of a treatment effect, a three-region rule might be

$$a_t = \begin{cases} \text{recommend}, & m_t \geq b_U(\log s_t), \\ \text{abandon}, & m_t \leq b_L(\log s_t), \\ \text{continue}, & b_L(\log s_t) < m_t < b_U(\log s_t), \end{cases}$$

with linear boundaries $b_U(u) = \alpha_U + \beta_U u$ and $b_L(u) = \alpha_L + \beta_L u$. Simulation or stochastic optimization chooses the boundary parameters to maximize expected utility subject to safety and error constraints. This resembles reinforcement learning in its sequential state–action–reward structure, though the trial design remains a prespecified statistical policy.

0.2.7 Confirmatory Phase III designs

A confirmatory trial usually compares a prespecified treatment with a control in a large randomized, multicenter study. A Bayesian final rule might declare success when

$$\mathbb{P}(\theta > \theta_0 \mid D_N) > \eta.$$

The posterior threshold η does not by itself determine a conventional Type I error. The induced false-positive probability is

$$\alpha_{\text{design}} = \mathbb{P}_{\theta=\theta_0}(\text{the complete design declares success}),$$

which depends on the prior, sample size, outcome model, stopping rules, adaptations, and all interim looks.

If success may be declared at looks $n_1 < \cdots < n_J = N$, then under the null

$$\alpha_{\text{design}} = \mathbb{P}_{\theta_0}\left(\bigcup_{j=1}^{J} \left\{\mathbb{P}(\theta > \theta_0 \mid D_{n_j}) > \eta_j\right\}\right).$$

This union probability is generally larger than the probability of crossing only the final boundary. Calibration must therefore simulate the full monitoring plan. Futility stopping may reduce expected sample size, but selection among several arms or doses introduces another multiplicity pathway and must also be included when controlling false success.

0.2.8 Seamless Phase II/III designs

A seamless design places exploratory selection and confirmatory comparison under one protocol. Stage 1 may investigate several doses; a prespecified transition rule selects one or more treatments for Stage 2 against control. Potential gains include shorter development time and reuse of eligible Stage 1 data, but the final analysis must account for the adaptation.

Let Z_1 and Z_2 be valid stagewise statistics whose null distributions are prespecified. A combination test may use

$$Z = w_1 Z_1 + w_2 Z_2, \qquad w_1^2 + w_2^2 = 1,$$

with critical values calibrated for the selection rule and dependence structure. Other methods use conditional error or closed testing. Simply pooling observations after selecting the most favorable Stage 1 arm can bias estimation and inflate Type I error. Treatment selection, sample-size adaptation, information timing, eligible Stage 1 data, and the final estimator must all be defined in advance.

0.2.9 Borrowing historical data

Historical controls can improve precision and reduce concurrent enrollment, which is attractive in rare diseases and settings with scarce participants. The central threat is nonexchangeability:

changes in eligibility, supportive care, endpoint assessment, calendar time, or data quality can make historical and current controls systematically different.

Let D_0 be historical data with likelihood $L_0(\theta)$ and D current data with likelihood $L(\theta)$. A fixed power prior is

$$\pi(\theta \mid D_0, a_0) \propto L_0(\theta)^{a_0}\pi_0(\theta), \qquad 0 \le a_0 \le 1. \tag{0.2.8}$$

Here $a_0 = 0$ gives no borrowing and $a_0 = 1$ gives full likelihood borrowing. In regular models, $a_0 n_0$ is a useful rough interpretation of historical effective sample size, but it is not universally exact.

For binary outcomes with historical x_0 of n_0 responses and baseline prior $p \sim \text{Beta}(a, b)$, the fixed power prior and current x of n responses give

$$p \mid D_0, D, a_0 \sim \text{Beta}(a + a_0 x_0 + x,\ b + a_0(n_0 - x_0) + n - x)\,.$$

Thus historical successes and failures contribute fractional counts.

If a_0 is unknown, its posterior must include the normalizing constant of the powered historical likelihood:

$$\pi(\theta, a_0 \mid D_0, D) \propto L(\theta)\frac{L_0(\theta)^{a_0}\pi_0(\theta)}{C(a_0)}\pi(a_0), \qquad C(a_0) = \int L_0(u)^{a_0}\pi_0(u)\ \mathrm{d}u.$$

Omitting $C(a_0)$ changes the posterior for the discount parameter. Hierarchical commensurate models offer another approach: historical and current parameters are related through a between-study distribution so conflict increases heterogeneity and weakens borrowing.

Dynamic borrowing reduces but cannot eliminate bias under drift. A design should prespecify comparability variables, cap borrowing, examine prior–data conflict, and simulate bias, interval coverage, Type I error, power, and effective sample size under agreement and clinically plausible disagreement. The current randomized control remains the strongest protection against unmeasured population differences.

0.2.10 Connection to AI-system evaluation

The same principles apply when an AI system is evaluated prospectively: performance targets must be clinically meaningful; cases are the sampling units; repeated looks and model revisions can inflate false-positive claims; adaptive allocation changes the observed case mix; and a Bayesian probability statement does not replace operating-characteristic evaluation. Unlike a simple Bernoulli response model, AI errors may be correlated within patients, sites, readers, or documents and may differ across subgroups. Trial-design ideas can guide evaluation, but the statistical model must reflect that dependence and the intended clinical use.

0.3 Mathematical Foundations of Large Language Models

For tokens $x_{1:T}$, an autoregressive language model uses the chain rule

$$p_\theta(x_{1:T}) = \prod_{t=1}^{T} p_\theta(x_t \mid x_{<t}), \tag{0.3.1}$$

and pretraining minimizes empirical negative log-likelihood,

$$\mathcal{L}_{\mathrm{NLL}}(\theta) = -\sum_{i=1}^{n}\sum_{t=1}^{T_i} \log p_\theta(x_{it} \mid x_{i,<t}).$$

Teacher forcing supplies the true preceding tokens during training; generation feeds selected or sampled outputs back into the context. Per-token cross-entropy is often summarized by perplexity $\exp\{\mathcal{L}_{\mathrm{NLL}}/N_{\mathrm{tok}}\}$, although perplexity does not directly measure clinical correctness or instruction following.

An embedding matrix maps tokens to $h_t^{(0)} = E[x_t] + p_t \in \mathbb{R}^{d_{\mathrm{model}}}$, where p_t supplies position information. A sequence model constructs contextual states, and a final linear map and softmax give

$$p_\theta(x_{t+1} = v \mid x_{\le t}) = \mathrm{softmax}(W_o h_t + b_o)_v.$$

0.3.1 Transformer architecture

A transformer alternates token mixing by attention with positionwise nonlinear processing. A common pre-normalized decoder block is

$$U^{(\ell)} = H^{(\ell)} + operatorname{MHA}\Big(\mathrm{LN}(H^{(\ell)})\Big),$$
$$H^{(\ell+1)} = U^{(\ell)} + operatorname{MLP}\Big(\mathrm{LN}(U^{(\ell)})\Big).$$

Residual paths improve information and gradient flow. A gated MLP may use

$$\mathrm{MLP}(h) = W_2[\sigma(W_g h) \odot (W_1 h)],$$

with SiLU or another activation σ.

An *encoder* uses bidirectional self-attention for representation or classification; a *decoder* uses causal self-attention for autoregressive generation; and an *encoder–decoder* adds cross-attention from decoder queries to encoder states. Learned absolute embeddings, sinusoidal features, relative biases, or rotary embeddings may encode position.

Proposition 0.3.1 (Permutation equivariance without position)**.** *Let P permute sequence positions and*

$$A(X) = \mathrm{softmax}_{\mathrm{row}}\left(\frac{XW_Q(XW_K)^\top}{\sqrt{d_k}}\right)XW_V.$$

Without positional features or a position-dependent mask, $A(PX) = PA(X)$.

Proof. Writing $Q = XW_Q$, $K = XW_K$, and $V = XW_V$, permutation yields PQ, PK, PV and the score matrix $P(QK^\top)P^\top$. Row-wise softmax commutes with simultaneous row and column permutation. Multiplication by PV then gives $PA(X)$. □

Thus attention alone recognizes content but not order; positional mechanisms break this symmetry in a controlled fashion.

0.3.2 Attention and its main variants

For queries Q, keys K, values V, and mask M, scaled dot-product attention is

$$\mathrm{Attn}(Q, K, V; M) = \mathrm{softmax}_{\mathrm{row}}\left(\frac{QK^\top}{\sqrt{d_k}} + M\right)V. \tag{0.3.2}$$

An allowed position has $M_{ij} = 0$ and a prohibited one has $M_{ij} = -\infty$. Each output is a convex combination of value vectors. Scaling stabilizes the logits: if query and key coordinates are independent, mean zero, and variance one, then

$$\mathrm{Var}(q^\top k) = d_k, \qquad \mathrm{Var}(q^\top k/\sqrt{d_k}) = 1.$$

Without scaling, high-dimensional scores can saturate softmax.

Multi-head attention learns different projections:

$$\begin{aligned} \mathrm{head}_r &= \mathrm{Attn}(XW_Q^{(r)}, XW_K^{(r)}, XW_V^{(r)}; M), \\ \mathrm{MHA}(X) &= \mathrm{Concat}(\mathrm{head}_1, \ldots, \mathrm{head}_h)W_O. \end{aligned}$$

Major variants include:

- **self-attention**, with Q, K, V from one sequence, and **cross-attention**, with queries and key–values from different sequences;
- **causal attention**, masking $j > i$, and **bidirectional attention**, permitting all non-padding interactions;
- **local/sliding-window attention**, restricting each token to a neighborhood, and structured **sparse attention**, adding selected blocks or global tokens;
- **multi-query attention**, sharing one key–value projection across query heads, and **grouped-query attention**, sharing within groups; both reduce autoregressive key–value-cache memory;
- **kernel or linear attention**, using a feature map so products can be reassociated, trading exact softmax behavior for improved scaling.

Full attention requires $O(n^2 d)$ pairwise arithmetic. Memory-efficient exact implementations need not store the full $n \times n$ matrix, but the arithmetic remains quadratic; a local window of width w costs about $O(nwd)$.

0.3.3 State-space sequence models

A continuous-time linear state-space model (SSM) is

$$\dot{h}(t) = Ah(t) + Bx(t), \qquad y(t) = Ch(t) + Dx(t).$$

Zero-order-hold discretization with step Δ gives

$$h_t = \bar{A}h_{t-1} + \bar{B}x_t, \qquad y_t = Ch_t + Dx_t, \tag{0.3.3}$$

where

$$\bar{A} = e^{\Delta A}, \qquad \bar{B} = \int_0^{\Delta} e^{(\Delta - s)A} B \, \mathrm{d}s.$$

If A is invertible, $\bar{B} = A^{-1}(e^{\Delta A} - I)B$.

Proposition 0.3.2 (Linear SSM as a convolution)**.** *For eqrefeq:discrete-ssm with $h_0 = 0$,*

$$y_t = Dx_t + \sum_{j=1}^{t} C\bar{A}^{t-j}\bar{B}x_j.$$

Thus a time-invariant linear SSM is a causal convolution with lag kernel $C\bar{A}^{\ell}\bar{B}$, plus the direct term.

Proof. Repeated substitution yields $h_t = \sum_{j=1}^{t} \bar{A}^{t-j}\bar{B}x_j$; substitute this into the output equation. □

The recurrence supports constant-size-state inference, while the convolution or an associative scan supports parallel training. Eigenvalues of $\bar{A}$ near the unit circle create slow-decaying memory; magnitudes above one can produce unstable states. Selective SSMs allow parameters such as input/output projections or step sizes to depend on the current token. This enables content-dependent retention but removes the fixed convolution-kernel view. Attention explicitly revisits prior tokens; an SSM compresses the prefix into a state. Hybrid models combine both.

0.3.4 Sentence embeddings and sentence transformers

A sentence embedding maps a variable-length sequence to $z = f_\theta(x_{1:T}) \in \mathbb{R}^d$. Given contextual token vectors, pooling may use a designated token, componentwise maxima, learned attention, or masked mean

$$z = \frac{\sum_{t=1}^{T} m_t h_t}{\sum_{t=1}^{T} m_t}.$$

Embeddings are often normalized and compared by cosine similarity $s(x, x') = \widehat{z}_x^{\top}\widehat{z}_{x'}$.

Proposition 0.3.3 (Cosine and Euclidean rankings)**.** *For unit vectors u, v, $\|u - v\|^2 = 2 - 2u^{\top}v$. Therefore decreasing cosine-similarity rank equals increasing Euclidean-distance rank.*

Proof. Expand the squared norm and use $\|u\| = \|v\| = 1$. □

A sentence transformer is commonly a shared-weight bi-encoder that embeds two sentences independently. For matched pairs and in-batch candidates, a contrastive objective is

$$\mathcal{L}_i = -\log \frac{\exp\{s(z_i, z_i^+)/\tau\}}{\sum_{j=1}^{B} \exp\{s(z_i, z_j^+)/\tau\}}, \tag{0.3.4}$$

where τ is a temperature. Random, in-batch, and hard negatives teach different distinctions; false negatives can corrupt the geometry.

Bi-encoders permit corpus embeddings to be precomputed and indexed. A new query needs one encoding plus nearest-neighbor search. A cross-encoder jointly processes each query–candidate pair and is often more accurate but much more expensive. Retrieval systems commonly use a bi-encoder followed by cross-encoder reranking. In clinical text, negation, temporality, dosage, and abbreviations can make nearby embeddings clinically incompatible, so proximity is candidate evidence rather than entailment.

0.3.5 Reinforcement-learning fine-tuning

Supervised fine-tuning imitates demonstrations. Reinforcement-learning fine-tuning optimizes a preference-derived or programmatic reward. A common RLHF workflow trains a supervised reference policy, collects response preferences, fits a reward model, optimizes a policy under a proximity constraint, and evaluates it independently of the training reward.

For preferred and dispreferred responses y^+, y^-, a Bradley–Terry reward loss is

$$\mathcal{L}_{\mathrm{RM}}(\phi) = -\log \sigma\{r_\phi(x, y^+) - r_\phi(x, y^-)\}.$$

It identifies reward differences, not an absolute scale. A KL-regularized policy objective is

$$\max_{\pi} \ \mathbb{E}_{x,y\sim\pi(\cdot|x)}[r(x,y)] - \beta \mathbb{E}_x[\mathrm{KL}\{\pi(\cdot \mid x) \| \pi_{\mathrm{ref}}(\cdot \mid x)\}]\,. \tag{0.3.5}$$

Theorem 0.3.1 (Optimal KL-regularized policy)**.** *For fixed x, finite response space, and $\beta > 0$, the maximizer supported by π_{ref} is*

$$\pi^*(y \mid x) = \frac{\pi_{\mathrm{ref}}(y \mid x) e^{r(x,y)/\beta}}{\sum_{y'} \pi_{\mathrm{ref}}(y' \mid x) e^{r(x,y')/\beta}}.$$

Proof. Add multiplier λ for $\sum_y \pi_y = 1$. Differentiating the objective gives

$$r_y - \beta\{\log(\pi_y/\pi_{\mathrm{ref},y}) + 1\} + \lambda = 0.$$

Hence $\pi_y \propto \pi_{\mathrm{ref},y} e^{r_y/\beta}$; normalization gives the claim. Strict concavity gives uniqueness on the supported simplex. □

Smaller β favors reward more strongly; larger β stays closer to the reference. Because responses cannot be enumerated, policy-gradient methods approximate this solution. PPO uses ratio $\rho_t(\theta) = \pi_\theta(a_t \mid s_t)/\pi_{\text{old}}(a_t \mid s_t)$ and the clipped surrogate

$$\mathcal{L}_{\text{PPO}}(\theta) = \mathbb{E}_t \left[\min\{\rho_t(\theta)\widehat{A}_t, \text{clip}(\rho_t(\theta), 1-\epsilon, 1+\epsilon)\widehat{A}_t\} \right].$$

Direct preference optimization (DPO) avoids a separate reward-model sampling loop during policy training and uses

$$\mathcal{L}_{\text{DPO}}(\theta) = -\log \sigma\left(\beta \left[\log \frac{\pi_\theta(y^+ \mid x)}{\pi_{\text{ref}}(y^+ \mid x)} - \log \frac{\pi_\theta(y^- \mid x)}{\pi_{\text{ref}}(y^- \mid x)} \right] \right).$$

Related methods use AI feedback, verifiable rewards, or group-relative baselines. Any reward remains a proxy: a model may learn verbosity, formatting, or loopholes rather than truthfulness. Biomedical use therefore requires held-out clinical evaluation, provenance checks, subgroup analysis, and human oversight beyond reward optimization.

0.4 Adapting Generative AI for Biomedical Research

A general-purpose language model may possess broad linguistic and reasoning ability without knowing a study's private documents, current data, required schema, or institutional rules. *Adaptation* supplies the context or behavior needed for a particular biomedical workflow. Three principal methods form an increasing ladder of intervention:

1. **Prompt engineering** changes the instructions and examples in the model input but leaves model weights unchanged;
2. **retrieval-augmented generation (RAG)** retrieves relevant external evidence and places it in the model context;
3. **supervised fine-tuning (SFT)** trains model parameters or adapters on curated input–output examples.

These methods solve different problems. Prompting is appropriate for task definition and output control, RAG for changing or private knowledge, and SFT for persistent task behavior that is demonstrated by enough high-quality data.

0.4.1 Models and access layers

The most specialized biomedical model is not automatically the best foundation for every biomedical system. General models may offer stronger reasoning, coding, multimodal processing, and tool use; domain-specific models may better represent clinical vocabulary; smaller open-weight models may be preferred for cost, latency, customization, or local control. Selection should therefore be based on task-specific evaluation rather than the model label alone.

There are three common access layers:

- a **web interface**, useful for interactive exploration but less suitable for automated, versioned processing;
- a **cloud API**, suitable for scalable and reproducible systems, with cost and data-governance requirements;
- **local deployment**, offering greater infrastructure and data control at the cost of hardware, engineering, monitoring, and maintenance.

No access layer is intrinsically safe for protected health information (PHI). The decision depends on contracts, data retention and training policies, encryption, access controls, auditability, regional requirements, and institutional approval. De-identification lowers some risks but does not by itself authorize use of a service.

For local inference, memory is often the primary constraint. A model with P parameters stored at b bits per parameter requires approximately

$$\frac{Pb}{8} \text{ bytes}$$

for weights alone. At 16-bit precision this is about 2 GB per billion parameters; actual inference additionally needs memory for activations, the key–value cache, runtime buffers, and framework overhead. Quantization reduces weight memory but can affect output quality and hardware compatibility.

0.4.2 Prompt structure and generation controls

An application should separate stable instructions from the instance-specific request. A *system instruction* defines the role, boundaries, terminology, missing-data behavior, and output contract. A *user query* supplies the current document or record and the task to perform. Keeping these layers separate makes the study procedure easier to version and audit.

A robust extraction prompt usually specifies:

1. the task and intended clinical meaning of each field;
2. the allowed evidence—for example, only the supplied note;
3. a schema, data types, controlled vocabularies, and date or unit formats;
4. explicit behavior for missing, conflicting, or uncertain information;
5. one or more representative examples when zero-shot instructions are insufficient;
6. a prohibition on unsupported inference and a requirement for source spans or citations when verification matters.

Example 0.4.1 (A typed extraction contract)**.** Suppose a note must be mapped to age, gender, primary diagnosis, medications, and admission date. A useful contract might require

```
{
  "patient_age": integer | null,
  "patient_gender": string | null,
  "primary_diagnosis": string | null,
  "medication_names": [string],
  "admission_date": "YYYY-MM-DD" | null,
  "evidence": {"field_name": "verbatim source span"}
}
```

The program should parse this output and validate it against a schema. A JSON- looking Markdown block, a misspelled key, or an age returned as free text is not equivalent to valid structured data. Failed validation should trigger repair or review, not silent coercion.

Zero-shot prompting supplies instructions only. One-shot and few-shot prompting also supply demonstrations. Examples can stabilize key names, types, and normalization choices, but they can also propagate mistakes or unrepresentative conventions. Demonstrations should therefore be curated, versioned, and tested on examples unlike those shown in the prompt.

If a model produces token logits $z_1, \ldots, z_K$, temperature sampling commonly uses

$$p_i(\tau) = \frac{\exp(z_i/\tau)}{\sum_{j=1}^{K} \exp(z_j/\tau)}, \qquad \tau > 0.$$

Smaller τ concentrates mass on high-logit tokens; larger τ produces more varied output. As $\tau \downarrow 0$, decoding approaches a choice among maximum-logit tokens. Low temperature can improve repeatability, but it does not make a false answer true, and complete reproducibility can still depend on the provider, model version, decoding implementation, and seed.

0.4.3 Hallucination and grounding

A hallucination is fluent but unsupported or false generated content, such as an invented citation, trial, dosage, patient attribute, or numerical result. The underlying model estimates likely token continuations; it is not a database of verified truth. It may also lack private data or information released after training. In biomedical work these are safety and scientific-validity risks, not merely stylistic defects.

Several controls are complementary:

- ground answers in supplied or retrieved evidence;
- require citations that point to exact source passages;
- use tools for database lookup, arithmetic, and code execution;
- distinguish “not found” from negative clinical evidence;
- validate types, ranges, terminology, and cross-field consistency;
- measure unsupported-claim and abstention behavior on held-out cases;

- keep human review for high-consequence decisions.

Prompting a model to "never hallucinate" is a policy statement, not a complete control mechanism.

0.4.4 Retrieval-augmented generation

RAG treats the model as taking an open-book examination. It connects an LLM to an indexed corpus so that only a small set of relevant evidence is placed in the context. This is useful when a corpus exceeds the model's context window, when sending every document is expensive, or when knowledge is private or changes frequently.

0.4.4.1 Indexing and inference workflows

An offline indexing workflow typically performs

$$\text{documents} \to \text{parse/OCR} \to \text{normalize} \to \text{chunk} \to \text{embed/index} \to \text{store metadata}.$$

At query time, the workflow is

$$\begin{aligned}\text{query} \to \text{retrieve candidates} \to \text{rerank} \to \text{construct grounded prompt} \\ \longrightarrow \text{generate} \to \text{verify and cite}.\end{aligned}$$

Metadata should include document identity, version, page or section, chunk boundaries, and access-control labels. Retrieval must enforce the same data permissions as the underlying source systems.

Clinical documents make parsing consequential. PDF text extraction can lose columns, headings, superscripts, and table structure; scanned pages require OCR; spreadsheets need sheet, row, and column context. Before indexing, a pipeline should assess extraction quality and preserve links to the original artifact.

0.4.4.2 Chunking

Chunking determines the unit of retrieval. Fixed token windows are simple but may split definitions from exceptions. Overlapping recursive windows preserve some local context at increased storage and redundancy. Semantic chunking uses topic changes, while document-aware chunking respects headings, pages, tables, and protocol sections. Model-directed or agentic chunking can identify logical units, but it introduces cost and another model-dependent step.

There is a basic tradeoff. Chunks that are too small omit necessary context; chunks that are too large dilute retrieval scores and consume the generation context. For clinical rules, parent–child retrieval is often useful: index a small clause for matching but return its enclosing criterion or section for interpretation. Tables should usually be serialized with headers repeated so that a retrieved row retains its meaning.

0.4.4.3 Sparse, dense, and hybrid retrieval

Sparse retrieval represents lexical occurrence. One conventional weight for term t in document d is

$$\mathrm{tfidf}(t,d) = \mathrm{tf}(t,d)\left[\log\left(\frac{N+1}{\mathrm{df}(t)+1}\right)+1\right],$$

where N is the number of documents and $\mathrm{df}(t)$ counts documents containing t. Methods such as BM25 improve length normalization and term- frequency saturation. Lexical retrieval is valuable for exact drug names, codes, rare genes, and quoted phrases.

Dense retrieval maps text to an embedding $v \in \mathbb{R}^d$ intended to represent semantic meaning. Common comparison functions are

$$\cos(q,v) = \frac{q^\mathsf{T} v}{\|q\|\,\|v\|}, \qquad q^\mathsf{T} v, \qquad \|q-v\|_2 .$$

Cosine similarity ignores vector magnitude; dot product does not unless vectors are normalized; Euclidean distance reverses the ordering convention because smaller is better. The embedding model and similarity metric must be matched.

Hybrid retrieval fuses sparse and dense scores or rankings. It is often more robust in biomedical corpora because a query can require both conceptual matching and exact identifiers. Knowledge-graph retrieval instead represents entities and relations explicitly, which can support multi-hop questions but requires reliable entity resolution and graph construction.

Exact nearest-neighbor search compares the query with every vector. Approximate indexes trade a small amount of recall for speed: inverted-file indexes search nearby clusters, graph indexes such as HNSW traverse a small-world graph, and product quantization compresses vectors. Index quality should be evaluated by retrieval recall and end-task performance, not speed alone.

0.4.4.4 Evaluating a RAG system

Evaluation should separate retrieval from generation. Useful retrieval metrics include recall@k, precision@k, mean reciprocal rank, and whether the gold evidence was retrieved. Generation evaluation asks whether the answer is correct, supported by retrieved evidence, complete, properly cited, and willing to abstain when evidence is absent. End-to-end accuracy alone cannot reveal whether failure arose from parsing, chunking, retrieval, context construction, or the generator.

0.4.5 Supervised fine-tuning and LoRA

SFT uses labeled input–output pairs (x_i, y_i) to modify model behavior. For autoregressive generation, training minimizes token-level negative log likelihood,

$$\mathcal{L}(\phi) = -\sum_i \sum_{t=1}^{|y_i|} \log p_\phi(y_{it} \mid x_i, y_{i,<t}),$$

where ϕ denotes the trainable parameters. Suitable applications include stable clinical coding behavior, specialized terminology, or complex output formats that prompting alone does not reliably produce. SFT is usually a poor substitute for RAG when the problem is simply access to changing factual knowledge.

Full fine-tuning updates all model weights and can be expensive. Low-rank adaptation (LoRA) freezes a weight matrix $W \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$ and learns

$$W' = W + \frac{\alpha}{r} BA, \qquad A \in \mathbb{R}^{r \times d_{\text{in}}}, \quad B \in \mathbb{R}^{d_{\text{out}} \times r}, \qquad r \ll \min(d_{\text{in}}, d_{\text{out}}).$$

The layer output becomes

$$h = W'x = Wx + \frac{\alpha}{r} BAx.$$

Instead of training $d_{\text{out}} d_{\text{in}}$ entries, the adapter trains $r(d_{\text{in}} + d_{\text{out}})$ entries. For a square $d \times d$ matrix this is $2dr$ rather than d^2. The rank r, scaling α, target modules, learning rate, batch size, epochs, and data mixture are validation choices, not harmless defaults.

Fine-tuning data should have clear provenance, consistent labels, appropriate consent and data-use authority, and train/validation/test separation that prevents patient or document leakage. Evaluation must check both desired task improvement and regressions in safety, calibration, general ability, and performance across clinically relevant subgroups. Adapters and base-model versions should be stored separately or merged only with explicit versioning.

0.4.6 Choosing an adaptation strategy

The smallest adequate intervention is usually easiest to validate:

Method	Use when	Primary risks
Prompting	Knowledge is already available and the main need is task definition, format, or demonstrations	Brittleness, prompt sensitivity, invalid structure, unsupported inference
RAG	Answers depend on private, large, attributable, or changing sources	Parsing and retrieval failure, stale indexes, access-control leakage, weak citations
SFT/LoRA	Persistent behavior is needed and many curated examples exist	Label error, memorization, data leakage, distribution shift, capability regression

Methods can be combined: an adapted model may still use RAG and strict output validation. The final choice should be driven by error analysis and a prespecified evaluation set, not by architectural novelty.

0.5 Agentic Systems and Tool Use

A plain language-model call maps a context to generated tokens. An *agentic system* places one or more model calls inside a stateful control loop that can select actions, call tools, observe

their results, revise its state, and decide when to stop. The surrounding software—not the model alone—enforces permissions, validates arguments, executes actions, records provenance, and applies termination rules.

At iteration t, a simple abstraction is

$$a_t = \pi_\phi(g, s_t, \mathcal{A}), \qquad o_t = \text{Execute}(a_t), \qquad s_{t+1} = \text{Update}(s_t, a_t, o_t),$$

where g is the goal, s_t is the current state, $\mathcal{A}$ is the allowed action set, a_t is a proposed action, and o_t is the observed result. The loop ends when a validated completion condition is met, when no safe action is available, or when a resource bound is reached.

0.5.1 Agent versus deterministic workflow

Agency is a spectrum rather than an all-or-nothing property:

- A **deterministic workflow** has a fixed sequence and explicit branches. The LLM may parse unstructured inputs at individual steps, but software controls the path. This is often the most auditable production choice.
- A **single autonomous agent** chooses among tools and determines its next step dynamically. It is flexible for exploratory tasks but harder to test exhaustively.
- A **graph with agentic nodes** fixes the allowed workflow graph while permitting model-based decisions within nodes. Typed interfaces constrain errors and make intermediate products inspectable.
- An **orchestrator–worker system** assigns subtasks to specialized workers and then combines their results. It can parallelize heterogeneous work, but introduces coordination, consistency, and provenance problems.

For a clinical workflow, autonomy should be granted only where variation is useful. A fixed pipeline is generally preferable when the process is known, regulated, and repeated. Agentic decisions are most useful when the path cannot be fully enumerated in advance, such as investigating conflicting evidence across a changing document collection.

0.5.2 ReAct and plan–execute patterns

In a ReAct-style loop, the system alternates between selecting one action and observing its result before selecting the next. This supports recovery because later choices can incorporate unexpected tool output. It may, however, take unnecessary steps or cycle without explicit progress checks.

In a plan–execute system, the model first proposes a sequence of steps and then executes it. Planning can reduce myopic action selection and make the intended trajectory reviewable, but a plan based on a false assumption can fail midway. A practical hybrid maintains a high-level plan while allowing bounded replanning after new evidence, rather than either wandering indefinitely or restarting the entire task after every deviation.

0.5.3 Function calling

Function calling separates a model's proposal from execution. A tool is defined by a name, purpose, parameter schema, output schema, permissions, and failure behavior. The model emits a structured request; the harness validates it, runs authorized code, and appends the result to the agent state.

Example 0.5.1 (A validated clinical calculation tool)**.** Suppose an agent may calculate a confidence interval but may not fabricate one. Its tool contract could be represented schematically as

```
name: binomial_confidence_interval
input:
  events: integer, minimum 0
  total: integer, minimum 1
  confidence_level: number in (0, 1)
  method: enum["exact", "wilson"]
output:
  estimate: number
  lower: number
  upper: number
  method: string
```

Before execution, ordinary code checks $0 \leq$ `events` $\leq$ `total` and validates every type and range. The calculator returns numbers and method metadata; the model may explain them, but it does not perform the underlying arithmetic itself. For consequential actions, the harness can require explicit human approval even after validation.

A bounded tool loop can be expressed as follows.

Algorithm 1 Validated agent loop

Require: goal g, initial context c, allowed tools $\mathcal{T}$, maximum turns K
$s \leftarrow \text{Initialize}(g, c)$
for $t = 1, \ldots, K$ **do**
  $r \leftarrow \text{Model}(s, \text{Schemas}(\mathcal{T}))$
  if r requests completion **then**
    validate required fields, citations, and completion criteria
    if valid **then**
      return r
    end if
  else
    validate tool name, arguments, permissions, and resource limits
    $o \leftarrow \text{ExecuteSafely}(r)$
    append the request, result, status, and provenance to s
  end if
end for
return an explicit incomplete status and the accumulated audit record

Tool descriptions should state what the tool does, not merely its Python signature. Ambiguous overlap between tools can cause unstable selection. Outputs should distinguish successful results, empty results, retryable errors, permission failures, and invalid requests. Tool calls should be idempotent where possible; actions with external side effects require stronger approval and duplicate-call protection.

0.5.4 Structured generation and semantic validation

An instruction such as "return JSON" is not the same as schema-constrained generation. Under ordinary prompting, the model may add Markdown fences, omit a required key, or return a number as a string. A JSON Schema defines allowed objects, required properties, types, enumerations, ranges, and whether additional fields are permitted.

When a model service supports native constrained decoding, the decoder tracks which token sequences can still lead to a schema-valid result and masks invalid continuations. If V_t^{valid} is the set of valid next tokens for the current partial output, the constrained distribution is

$$\widetilde{p}(v \mid c_t) = \frac{p(v \mid c_t)\mathbb{1}\{v \in V_t^{\text{valid}}\}}{\sum_{u \in V_t^{\text{valid}}} p(u \mid c_t)}.$$

This can guarantee syntactic structure when the schema and decoder are supported. It does not guarantee that a value is scientifically correct. A schema-valid p-value can still come from the wrong model, dataset, population, or contrast.

Clinical extraction contracts should therefore carry more than a value. A useful field representation includes

```
{
  "value": ...,
  "source_document": "...",
  "source_section": "...",
  "evidence": "verbatim supporting text",
  "status": "found | missing | conditional | conflicting"
}
```

The status prevents the agent from prematurely resolving language such as "use multiple imputation if missingness exceeds 10%; otherwise use observed cases." Both branches and their condition must be preserved until the data determine which applies. If a methods section specifies ANCOVA while a table shell specifies MMRM, the correct structured result is a conflict requiring adjudication, not a silent choice.

0.5.4.1 JSON, schema, and structured output are different

JSON is a notation that a parser can read. JSON Schema is a producer–consumer contract specifying the expected keys, types, cardinalities, allowed values, and extra-field policy. Structured output is the generation mechanism used to produce an instance of that contract. These form a ladder:

$$\text{free text} \longrightarrow \text{requested format} \longrightarrow \text{valid JSON} \longrightarrow \text{schema-constrained record}.$$

Moving upward reduces formatting uncertainty, but none of these steps alone establishes factual correctness.

A schema lets the producer promise stable field names, required fields, and types, while the consumer can perform predictable lookup and type-safe operations. Disallowing unspecified properties also makes unexpected content detectable. Schema design is nevertheless a scientific choice: a perfectly enforced but conceptually wrong schema creates consistently wrong records.

0.5.4.2 Begin with the unit of record and downstream use

The first question is what one object represents: a trial, arm, endpoint, eligibility criterion, analysis, or estimand. This is analogous to defining what one row means in a statistical dataset. A sentence stating that a trial has two co-primary endpoints should normally produce two endpoint records, not one string containing both names.

The downstream use determines the appropriate representation:

- summarizing one protocol may prioritize readability and rich source text;
- comparing many trials requires stable fields, controlled vocabulary, and explicit missingness;
- quantitative analysis additionally requires numeric types, units, estimands, and deterministic validation rules.

Copying document headings into JSON does not necessarily model the objects and relationships needed by the workflow.

Shared context should be stored once, while repeatable objects belong in arrays:

```
{
  "trial_id": "NCT01866319",
  "document_id": "KEYNOTE006_report_v1",
  "treatment_arms": [...],
  "endpoints": [ {...}, {...} ],
  "extraction_metadata": {...}
}
```

Using a scalar '`primary_endpoint`' silently assumes there can be only one. Cardinality is therefore a scientific assumption encoded in the data model, not a cosmetic programming decision.

0.5.4.3 Types, nesting, and controlled vocabularies

Types determine which downstream operations are valid. The string '`"18 years"`' requires reparsing before numerical comparison, whereas

```
"minimum_age": {"value": 18, "unit": "years"}
```

supports range checks and explicit unit conversion. A Boolean should be represented as '`true`' or '`false`', not one of many strings such as "Yes," "Y," or "TRUE." If the document is silent, null plus an information status is different from false.

Nesting preserves relationships. Instead of storing '`"endpoint": "PFS at 24 months by central review"`', represent the endpoint name, timeframe, unit, and assessment method as related fields. This prevents downstream code from repeatedly parsing an opaque phrase and allows a missing component to be represented independently.

Controlled vocabularies make cross-study comparison possible. For example, endpoint role might be restricted to '`primary`', '`secondary`', '`exploratory`', '`safety`', '`other`', and '`unclear`'. Such an enumeration is useful only if it fits the scientific task. Forcing an unsupported category is worse than preserving an unclear status.

0.5.4.4 Missingness and uncertainty are data

A required field need not be known. Requiring a nullable field means every record must address the question, even when the answer is unavailable. A single null, however, can erase clinically relevant distinctions. Useful statuses include:

Status	Meaning	Typical action
reported	The source states the value clearly	Validate and use
not reported	The searched source does not provide the value	Search another source
ambiguous	More than one interpretation is plausible	Request review
not applicable	The field does not apply to this design	Exclude appropriately
conflicting	Sources or passages disagree	Reconcile evidence

Collapsing these states into null loses decision-relevant information. A raw model confidence such as 0.95 is not a substitute unless its meaning and calibration have been defined and evaluated for the task.

0.5.4.5 A worked endpoint record

An endpoint record should preserve identity, scientific meaning, uncertainty, and evidence. One possible design is:

```
{
  "endpoint_id": "ep_01",
  "endpoint_role": "primary",
  "name_as_reported": "progression-free survival",
  "normalized_concept": "PFS",
  "definition": {"value": null, "status": "not_reported"},
  "timeframe": {
    "value": null, "unit": null, "status": "not_reported"
  },
  "assessment_method": {
```

```
    "value": "blinded_central_review", "status": "reported"
  },
  "analysis_population": {
    "value": "all_randomized", "status": "reported"
  },
  "treatment_contrast": {"value": null, "status": "not_reported"},
  "source": {
    "document_id": "KEYNOTE006_report_v1",
    "section": "Methods",
    "printed_page": 3,
    "source_text": "...supporting passage..."
  },
  "interpretation_notes": []
}
```

The original wording and normalized value are both retained. Normalization supports comparison but must not erase the evidence it summarizes. Page number alone is weak provenance because PDF and printed pagination may differ, versions can move content, and one page contains many claims. Exact supporting text, section, document identifier, and version should travel with the value.

0.5.4.6 Extraction, normalization, and inference

These are distinct transformations:

$$\underbrace{\text{“all randomized participants”}}_{\text{extracted}} \longrightarrow \underbrace{\texttt{all_randomized}}_{\text{normalized}} \longrightarrow \underbrace{\text{“ITT-like”}}_{\text{inferred}}.$$

The first records source language, the second maps it to a common label, and the third applies domain interpretation. Combining all three into ‘`ITT`’ makes an error difficult to localize and may overstate what the source says. The record should distinguish them and state whether inference was permitted.

0.5.4.7 Layered validation

Reliable extraction is a workflow with a prompt, a schema, and validators. It should ask separate questions:

1. **Structural:** does the object satisfy required keys, types, cardinalities, enums, and extra-field policy?
2. **Semantic:** are values and relationships plausible, such as nonnegative age and lower bound no greater than upper bound?
3. **Referential:** do a contrast’s treatment arms exist, and do identifiers resolve to the same trial and document version?
4. **Clinical:** is PFS classified as time-to-event, and do the chosen fields preserve endpoint meaning?

5. **Evidential:** does the cited passage explicitly support the value, is context omitted, and do other passages conflict?

Deterministic code should enforce checks such as

$$\texttt{status=not_reported} \Rightarrow \texttt{value=null}, \qquad \texttt{lower_bound} \leq \texttt{upper_bound}.$$

Language-based evidence review is a separate step, ideally with the claim and source passage presented together. Human review remains appropriate for ambiguity, conflicts, and consequential scientific mappings.

Eligibility criteria add further logical structure. One sentence can encode multiple linked rule records containing inclusion/exclusion type, clinical domain, concept, operator, value and unit, time window, negation, Boolean connective, and exception. Flattening the entire sentence into one criterion string prevents reliable computation of nested AND/OR logic.

0.5.5 Model Context Protocol

Ordinary function calling is application-specific: each team may invent its own schemas, transports, authentication, and result formats. Model Context Protocol (MCP) provides a standardized way for an AI host to discover and call capabilities supplied by external servers.

Conceptually, an MCP deployment contains:

- a **host**, the agent application that manages the model, orchestration, permissions, and user interaction;
- a **client**, maintained by the host for communicating with an MCP server;
- a **server**, which exposes tools or resources backed by code, databases, or other services.

MCP standardizes the interface, not the correctness or safety of the underlying operation. Authentication, least-privilege authorization, input validation, network isolation, logging, version pinning, and data-governance review remain necessary. A server that can read a clinical database should expose only the minimum queries and records required for the approved task.

0.5.6 Skills and progressive context loading

A skill is a reusable package of task instructions, domain rules, examples, schemas, or workflow guidance. Rather than place an entire manual in every prompt, an agent can discover the relevant skill and load only the portions needed for the current task. This progressive disclosure conserves context and reduces interference from irrelevant instructions.

Skills and tools play different roles. A skill tells the system *how* to perform a process; a tool gives it an executable capability. For example, a protocol-extraction skill may define the endpoint schema, cross-reference resolution procedure, and review checklist, while an MCP tool fetches protocol pages from an approved repository. Both should be versioned because a change to instructions can alter outputs as materially as a code change.

0.5.7 Worked workflow: SAP to primary analysis

Consider a study whose SAP defines the primary endpoint as change from baseline in HbA1c at Week 24, the full analysis set (FAS) as the primary population, and the model

$$Y_i = \beta_0 + \beta_1 \mathbb{1}\{A_i = 1\} + \beta_2 B_i + \varepsilon_i, \qquad \mathbb{E}(\varepsilon_i \mid A_i, B_i) = 0, \tag{0.5.1}$$

where Y_i is Week 24 change, A_i is randomized treatment, and B_i is baseline HbA1c. The treatment contrast is β_1, adjusted for baseline. The agentic workflow should be decomposed before implementation:

1. extract endpoint, visit, population, model, covariates, contrast, missing-data rule, and source evidence from the SAP;
2. validate that required subject-level and efficacy variables exist;
3. verify that baseline and Week 24 records can be linked and that the endpoint derivation matches the SAP;
4. construct the FAS from its operational definition and reconcile it with population flags;
5. run tested ANCOVA code only after every required QC gate passes;
6. review sample sizes by arm, exclusions and missingness, coefficient estimates, confidence intervals, p-values, warnings, diagnostics, and provenance.

The LLM is useful in Step 1 because SAP language varies and information can be distributed across sections. Steps 2–5 are principally deterministic. The agent may decide when to invoke them, but tested code determines whether a column exists, whether treatment values are allowed, whether endpoint subjects belong to the FAS, and how the regression is fit.

For two treatment groups, a minimal analysis contract could require

$$\{\texttt{USUBJID}, \texttt{TRT01P}, \texttt{BASE}, \texttt{CHG}, \texttt{AVISIT}, \texttt{FASFL}\}.$$

Semantic checks go beyond column presence: subject identifiers should be unique at the expected level; '`CHG`' should reconcile with post-baseline minus baseline values; Week 24 selection and visit windows should match the SAP; treatment categories should match randomization; and the fitted sample should equal the reported analysis population after missing-data rules.

The final artifact should bind together the SAP contract, document citations, dataset and code versions, QC table, model specification, software output, diagnostics, and reviewer status. A numerical result without this provenance is not an analysis-ready deliverable.

0.5.8 Hooks as deterministic enforcement

A *hook* is an event-triggered action run automatically by the harness before or after an operation. A skill says how an agent should perform a task; a hook enforces what the system

must do regardless of the model's choice. This distinction converts "remember to check" into "execution is impossible until the check passes."

For the ANCOVA workflow, a pre-tool hook can require:

- a complete, schema-valid SAP contract with no unresolved critical conflicts;
- presence and validity of required variables and treatment values;
- availability of the Week 24 endpoint under the specified visit rule;
- a reproducible FAS definition and agreement between endpoint subjects and the population;
- approved data, code, and analysis-contract versions.

The hook returns a structured pass/fail result with failed checks and evidence. Only a pass authorizes the ANCOVA tool. The model cannot override the gate by arguing that the analysis is probably safe.

Other useful hook points include:

- **session start:** load the approved protocol, SAP, versions, and study-specific permissions;
- **user input:** detect patient identifiers or prohibited data;
- **before tool use:** enforce authorization, protected paths, network destinations, QC prerequisites, and approval requirements;
- **after tool use:** validate identifiers, schemas, numerical ranges, and provenance records;
- **file change:** rerun statistical tests or QC when analysis code or derived data change;
- **completion:** require evidence tables, resolved failures, citations, and reviewer approval before reporting a complete status.

Hooks should themselves be small, deterministic, tested, fail closed for high-consequence operations, and emit auditable reasons. A hook that depends on an unconstrained LLM judgment is guidance disguised as enforcement.

0.5.9 Context and memory management

Agent contexts grow as prompts, plans, tool results, and intermediate drafts accumulate. More context is not always better: irrelevant material can dilute attention, create contradictory instructions, increase latency and cost, and make provenance unclear. Context management should therefore distinguish:

- **working state**, the compact information required for the next decision;
- **authoritative evidence**, retrieved on demand with citations;

- **episodic history**, prior actions and results used for recovery and audit;
- **durable memory**, only facts explicitly approved for reuse across tasks.

Long tool outputs can be stored outside the prompt and referenced by identifiers. Summaries should retain source links and critical exceptions, and a later step should be able to retrieve the original rather than treat a lossy summary as primary evidence. PHI should not be written into durable memory unless the storage and intended reuse are specifically authorized.

0.5.10 Reliability controls for agents

Agentic loops amplify both capability and error. Useful controls include:

1. set maximum turns, wall-clock time, token budgets, tool-call counts, and monetary limits;
2. define typed inputs and outputs for every node and validate them at runtime;
3. expose only necessary tools and require approval for consequential or irreversible actions;
4. maintain per-step logs of model version, prompts, tool arguments, outputs, errors, citations, and status;
5. use deterministic software for arithmetic, date logic, database constraints, and statistical computation;
6. retry only errors likely to be transient, and cap retries;
7. return an explicit incomplete or uncertain status rather than forcing an answer when evidence or authority is missing.

Asking the same model to review its own answer is not an independent check. It may repeat or rationalize the same error because the draft anchors the review. A verifier is more useful when it receives explicit criteria and primary evidence, tests claims rather than tone, and can reject the answer. Independence can be strengthened with deterministic validation, a separately prompted or separately trained model, or human review. Even then, correlated failures are possible, so evaluation must measure the complete system rather than assume that adding another agent guarantees correctness.

Responsibility remains distributed across the system: the model interprets context and proposes actions; tools retrieve, validate, and compute; skills encode reusable procedures; state preserves artifacts and decisions; controls enforce gates and permissions; the harness coordinates execution and logging; and qualified humans resolve ambiguity and approve consequential decisions. Reliability comes from these layers working together, not from assigning every responsibility to a more capable model.

0.6 Eligibility Agents

Clinical-trial eligibility criteria sit at the boundary between scientific design and trial operations. Operationally, they protect participants and determine whether sites can recruit enough people. Scientifically, they define the population to which the treatment effect applies. A system that reverses a negation, drops a time window, or changes "at most one" to "none" does not merely make a clerical error: it changes the enrolled population and therefore the estimand.

Suppose a trial estimates

$$\tau_{\mathcal{E}} = \mathbb{E}[Y(1) - Y(0) \mid X \in \mathcal{E}]\,,$$

where $\mathcal{E}$ is the population satisfying the protocol's eligibility rules. If an automated screen actually applies an erroneous rule set $\widetilde{\mathcal{E}}$, its data target $\tau_{\widetilde{\mathcal{E}}}$ instead. The two quantities need not agree, especially when the mistaken criteria are treatment-history, disease-severity, or prognostic-factor restrictions. The same issue affects transportability and external controls: borrowing data is credible only when eligibility and baseline covariates define sufficiently comparable populations.

0.6.1 From prose to atomic computable rules

A single protocol bullet often contains several logically different rules. As a running example, consider a melanoma trial requiring patients to be at least 18 years old, to have unresectable stage III or IV melanoma, ECOG performance status 0 or 1, known BRAF V600 status, no prior ipilimumab, and at most one previous systemic therapy. The useful unit of extraction is not the sentence but the *atomic rule*:

$$\begin{aligned}
r_1(x) &= \mathbb{1}\{\text{age}(x) \geq 18\},\\
r_2(x) &= \mathbb{1}\{\text{stage}(x) \in \{\text{III}, \text{IV}\} \text{ and disease is unresectable}\},\\
r_3(x) &= \mathbb{1}\{\text{ECOG}(x) \in \{0, 1\}\},\\
r_4(x) &= \mathbb{1}\{\text{a BRAF V600 result is observed}\},\\
r_5(x) &= \mathbb{1}\{\text{no prior ipilimumab exposure}\},\\
r_6(x) &= \mathbb{1}\{\#\text{ prior systemic therapies} \leq 1\}.
\end{aligned}$$

If all are inclusion conditions, the computable eligibility indicator is

$$e(x) = \prod_{j=1}^{6} r_j(x).$$

This Boolean notation is deliberately incomplete until time and evidence are specified. For example, ECOG may have to be measured within 14 days before treatment, and treatment exposure must be assessed before enrollment. In practice the truth value is often three- or four-valued rather than Boolean: met, not met, unknown, or not applicable. Treating unknown as false silently turns missing data into ineligibility.

An extraction record should preserve both the source wording and a normalized meaning. Useful fields include

Field	Purpose
criterion identifier	Stable reference for review and downstream decisions.
polarity and atomic text	Inclusion versus exclusion, plus the exact clinical rule.
domain and variable	Clinical meaning, such as performance status/ECOG.
operator, value, and unit	The computable constraint, such as membership in $\{0, 1\}$.
time window	When the fact must hold relative to screening or treatment.
evidence span	Document, section, page, and exact supporting text.
computability	Direct, mapped, NLP required, clinical judgment, or unavailable.
uncertainty and review status	Ambiguous, conflicting, accepted, or requiring review.

The distinction between source and normalization prevents a common error: "known BRAF V600 status" means that a result is available, not that the mutation is positive. Similarly, "ipilimumab-naive" is a negated exposure, and "up to one" permits both zero and one.

0.6.1.1 Logic, time, and evidence

Relations are as important as entities. Extracting the tokens *ipilimumab*, *systemic therapy*, and *one* is insufficient unless the representation links

$$\neg\,\text{prior_ipilimumab}(x) \quad \wedge \quad \text{count}\{\text{prior systemic therapies of } x\} \leq 1.$$

AND, OR, NOT, counts, exceptions, and temporal relations should be stored explicitly, preferably as an abstract syntax tree or directed graph. That intermediate representation separates clinical meaning from local execution. It can later be translated to an OMOP cohort definition, SQL, or patient-note review without repeatedly reinterpreting the prose.

Computability is itself an output. A date of birth may support a direct age check; a laboratory value may first require concept and unit mapping; a cancer stage may require extraction from notes; "clinically significant" may require expert judgment; and an external treatment history may be unavailable. The label tells the agent whether to execute, map, retrieve, ask, or defer. Uncertainty should likewise distinguish *not reported*, *ambiguous*, *not applicable*, and *conflicting*; one null value loses the action implied by each state.

0.6.2 The representation lineage before large language models

Modern eligibility agents inherit a sequence of earlier design ideas:

- **ERGO** separated human-readable criteria from an intermediate clinical representation and its executable Boolean query. Its lasting principle is to define meaning before choosing database code.
- **EliXR** treated eligibility language as recurring semantic patterns with roles such as finding, modifier, stage, value, and polarity.
- **EliIE** decomposed extraction into entity recognition, negation detection, relation extraction, and normalization to a common data model. These subtasks remain present in LLM systems as prompts, tools, or validators.
- **Chia** represented entities, attributes, and their relations as directed graphs, making nested eligibility logic recoverable rather than leaving it implicit in a list of spans.
- **Criteria2Query** connected the semantic representation to OMOP concept sets and reviewable cohort queries. This exposed choices that text extraction alone cannot settle: which vocabulary concept, data table, observation window, and missing-data rule should be used?

For example, implementing "known BRAF V600 status" may require a reviewed concept set for variant tests, a search of measurement and molecular-report sources, and an existence query before enrollment. An executable approximation is

```
EXISTS (
  SELECT 1
  FROM measurement m
  WHERE m.person_id = p.person_id
    AND m.measurement_concept_id IN (<reviewed BRAF V600 test
       concepts>)
    AND m.measurement_date <= index_date
    AND m.value_as_concept_id IS NOT NULL
)
```

This query checks that status is observed, not that the result is positive. The concept set and source table are study- and database-specific artifacts that require review.

0.6.3 LLM extraction and query generation

Large language models replace some task-specific interfaces with a general instruction-following interface, but preprocessing and representation remain part of the method. An AutoCriteria-style workflow is:

1. identify inclusion and exclusion sections and preserve criterion identifiers;
2. split long text at sentence boundaries into manageable chunks;
3. prompt for attributes, values, modifiers, types, temporal phrases, relations, and exact evidence spans using a fixed output schema;

4. merge, deduplicate, and clean outputs;

5. validate a calibrated sample with domain experts and revise the extraction policy.

Domain knowledge belongs in the schema. Oncology emphasizes tumor characteristics, biomarkers, and treatment history; neurodegenerative disease may require cognitive scales and imaging; liver disease adds fibrosis and imaging measures. A general model does not eliminate these domain-specific types.

C2Q-style systems then assign distinct responsibilities to distinct steps: concept extraction, concept-set construction, SQL generation, explanation, and human editing. This decomposition matters because a fluent explanation is not evidence that the selected concept identifiers or query logic are correct. Each step should have its own input contract and evaluation target.

0.6.4 Patient-to-trial matching

Query generation works naturally for structured records, whereas oncology history is longitudinal and often buried in notes. A recruitment agent must therefore combine retrieval with criterion-level clinical reasoning. A TrialGPT-style workflow has three stages:

1. **Retrieval:** derive prioritized conditions from the patient summary, run sparse and biomedical dense retrieval, and fuse their rankings to reduce thousands of trials to a candidate set.

2. **Matching:** compare the patient evidence against every atomic inclusion and exclusion rule. Return a label, rationale, and evidence spans for each comparison.

3. **Ranking:** aggregate criterion-level decisions into a trial-level score while keeping the score traceable to its local evidence.

The labels must respect criterion polarity. For an inclusion rule, useful states are included, not included, insufficient information, and not applicable. For an exclusion rule, they are excluded, not excluded, insufficient information, and not applicable. A patient with stage IV melanoma, ECOG 1, documented ipilimumab exposure, and no BRAF report would be included on disease and performance status, excluded by the ipilimumab-naive rule, and unknown on BRAF evidence. Text similarity alone cannot produce this decision.

One transparent trial score uses hard vetoes before soft ranking. Let I_t and X_t be the inclusion and exclusion criteria for trial t. Define

$$V_t = \mathbb{1}\{\text{some } j \in I_t \text{ is not met, or some } j \in X_t \text{ is met}\}.$$

If $V_t = 1$, the trial is not currently eligible. Otherwise a review-priority score can be

$$S_t = \frac{\sum_j w_j m_{tj}}{\sum_j w_j} - \lambda \frac{\sum_j w_j u_{tj}}{\sum_j w_j},$$

where m_{tj} marks compatible criteria, u_{tj} marks unresolved criteria, and w_j reflects clinical importance. This is only a prioritization rule, not a probability of eligibility, unless it is

calibrated against labeled outcomes. The interface should display vetoes and unknowns rather than only S_t.

0.6.5 A controlled eligibility-agent loop

A reliable agent is a loop with narrow responsibilities:

$$\text{plan} \longrightarrow \text{extract} \longrightarrow \text{normalize} \longrightarrow \text{validate} \longrightarrow \text{review}.$$

On validation failure it revises the relevant artifact, not the clinical standard. Useful tools include a source parser, vocabulary lookup, unit converter, schema validator, evidence checker, database query runner, and a human-review queue. Code should verify arithmetic, units, dates, operators, and schemas; model judgment is reserved for language and contextual interpretation.

Validation proceeds in layers:

1. **Structural:** required fields, types, enumerations, and graph references satisfy the schema.
2. **Semantic:** concepts, units, operators, polarity, count constraints, and temporal order are clinically plausible.
3. **Evidential:** every extracted claim is entailed by a cited source span; normalized claims remain faithful to it.
4. **Clinical:** ambiguity, conflict, and judgment-dependent rules are routed to a qualified reviewer.

Valid JSON can still reverse a negation or invent a threshold, so structural validation alone is inadequate. Provenance should travel with each claim: source document and version, section, page, exact text, extraction run, mapping version, and reviewer decision. This allows an error to be localized to extraction, normalization, inference, or missing evidence.

Example 0.6.1 (Evidence-linked review)**.** Suppose the source says "known BRAF V600 status," but the model outputs `operator = equals`, `value = positive`, and no evidence. The schema validator may pass. Semantic validation fails because observed status was changed into positivity, and evidential validation fails because no source span supports the claim. The corrected object represents existence of a reported result and sends the actual positive/negative value to a separate field if the record supplies it.

Human-in-the-loop review should be targeted rather than a manual repetition of the whole workflow. Appropriate triggers include vague language such as "clinically significant," missing molecular results, conflicting treatment histories, inaccessible external records, low-confidence vocabulary mappings, and any unsupported hard exclusion. The reviewer sees the atomic rule, patient evidence, model decision, and failed checks together.

0.6.6 Evaluation and statistical consequences

Entity-level precision, recall, and F_1 are useful but do not fully measure an eligibility agent. A missed main criterion may be more consequential than several correctly extracted modifiers. Evaluation should therefore include:

- exact atomic-rule recovery, including negation, counts, OR groups, exceptions, units, and time windows;
- evidence-span precision and whether every normalized claim is supported;
- criterion-level matching accuracy with results stratified by inclusion/exclusion and known/unknown status;
- retrieval recall at the number of trials a reviewer can inspect;
- trial-level ranking and calibration, plus screening time and reviewer disagreement;
- error rates across demographic and clinical subgroups, sites, note styles, and missingness patterns.

Subgroup analysis is essential. If M denotes an incorrect match decision and G a patient subgroup, aggregate accuracy can conceal

$$\mathbb{P}(M = 1 \mid G = g_1) \neq \mathbb{P}(M = 1 \mid G = g_2).$$

Differential matching error changes who is approached or enrolled, creating selection bias and potentially changing the observed effect. Prospective monitoring should therefore track eligibility yields, reasons for exclusion, unknown rates, overrides, and downstream enrollment by subgroup and site.

Finally, the extracted rule object can support more than recruitment. It can compare populations across protocols, estimate feasibility in local data, identify restrictive criteria during protocol design, and assess whether an external cohort is similar enough for transport or borrowing. Reuse is safe only when the same evidence, versioning, uncertainty, and validation metadata travel with the object.

0.7 Schedule-of-Activities Agents

Eligibility criteria answer who may enroll; the schedule of activities (SOA) answers what should happen and when. Together they convert a scientific design into operational rules. The SOA may prescribe visits, treatment, laboratory tests, imaging, questionnaires, safety follow-up, and conditional assessments. Its apparent table of marks is not the complete specification: column headers, merged cells, footnotes, surrounding prose, protocol amendments, and arm- or phase-specific exceptions all contribute meaning.

The collection pathway should remain explicit:

$$\begin{aligned}\text{protocol plan} &\longrightarrow \text{site or vendor assessment} \longrightarrow \text{source record} \\ &\longrightarrow \text{eCRF/transfer} \longrightarrow \text{integrated trial data} \longrightarrow \text{analysis dataset}.\end{aligned}$$

A case report form (CRF) defines fields to be collected; an electronic CRF is the implemented form; and electronic data capture (EDC) is the system that hosts and manages those forms. Other systems—central laboratories, imaging vendors, and electronic patient-reported outcome platforms—may transfer data directly. Participant identifiers, dates, controlled terminology, and units must be reconciled when these sources are integrated.

0.7.1 Planned events are not observed measurements

An SOA describes planned requiredness. An observed-data record describes what actually happened. Imaging scheduled at Week 12 may occur at Week 13, a pre-dose laboratory test may be drawn after dosing, and a scheduled questionnaire may have no response. A robust data model therefore keeps separate variables for

- the nominal event and its allowed window;
- whether the event was required for this participant;
- whether it was performed;
- the actual date and time, result, unit, and method;
- if absent, the reason not performed;
- the source record and data-processing lineage.

Equating "scheduled" with "observed" masks protocol deviations and creates incorrect assumptions about missing data.

For a planned event j, let T_j^{nom} be its nominal time and $[L_j, U_j]$ its window relative to an anchor. If its actual time is T_j^{obs}, a simple timing classification is

$$D_j = \begin{cases} \text{missing}, & T_j^{\text{obs}} \text{ is absent}, \\ \text{early}, & T_j^{\text{obs}} < L_j, \\ \text{in window}, & L_j \le T_j^{\text{obs}} \le U_j, \\ \text{late}, & T_j^{\text{obs}} > U_j. \end{cases}$$

This classification is valid only after determining that the event was applicable and required. A conditional event whose trigger is false is not a missing required event.

0.7.2 A computable event representation

The extraction target is a scheduled event with rules and evidence, not merely a marked cell. A useful event record contains:

Field	Interpretation
event and activity identifiers	A stable occurrence and the clinical activity to perform.

timepoint, phase, and arm	Visit context and the population to which the event applies.
requiredness	Required, optional, or conditional, with the condition retained.
anchor, offset, and unit	The reference event and nominal displacement from it.
early/late window	Allowed deviation from nominal time, including whether boundaries are inclusive.
recurrence and stopping rule	Interval, number or end boundary, and transition to later schedules.
modalities and exceptions	Allowed methods and overrides such as "before dosing."
source evidence	Table, row, column-header path, cell symbol, footnote, page, and document version.
uncertainty and review	Specified, not reported, ambiguous, conflicting, or not applicable.

The full context of a cell is hierarchical. A chemistry mark beneath *Treatment / Cycle 2 / Day 1* cannot safely be represented as "Day 1" alone. Likewise, a footnote marker must be resolved within the correct table and protocol version. The original wording and its normalized meaning should travel together: for example, retain "Q6W" while normalizing its interval to six weeks.

0.7.2.1 Anchors and windows

Timing requires an anchor, offset, unit, and possibly asymmetric tolerances. For an anchor time A, offset d, early allowance w_-, and late allowance w_+,

$$T^{\text{nom}} = A + d, \qquad \mathcal{W} = [A + d - w_-,\ A + d + w_+].$$

Thus "Day 14 ±3 days after first dose" produces an allowed interval from Day 11 through Day 17 relative to the first-dose timestamp. Substituting the randomization date as A yields correct arithmetic for the wrong clinical rule. Anchor identity is therefore a semantic validation target, not a minor metadata field.

Different activities may use different clocks. Treatment can occur every three weeks while tumor imaging follows a study-week schedule. Treatment discontinuation also need not imply study discontinuation: a participant may stop drug, enter safety follow-up, and remain under survival follow-up. These transitions should be represented as distinct state and stopping rules.

0.7.2.2 Expanding recurrence rules

An LLM may interpret a recurrence phrase, but deterministic code should expand and check it. For start time s, interval $q > 0$, and inclusive endpoint e, the nominal sequence is

$$\mathcal{T}(s, q, e) = \{s + kq : k = 0, 1, \ldots, \lfloor (e - s)/q \rfloor\}.$$

For example, imaging every six weeks from Week 12 through Week 48 gives

$$\mathcal{T}(12, 6, 48) = \{12, 18, 24, 30, 36, 42, 48\}.$$

If the protocol then changes to every 12 weeks, the next rule starts at Week 60, not Week 48, unless the source explicitly duplicates the boundary event. A piecewise schedule can be written

$$\mathcal{T} = \mathcal{T}(12, 6, 48) \cup \{48 + 12k : k = 1, 2, \ldots\},$$

with the infinite second sequence truncated by a participant-level stopping condition such as progression, withdrawal, death, or study closure.

Example 0.7.1 (Detecting an unsupported schedule)**.** Suppose a published report says "Week 12, every six weeks through Week 48, then every 12 weeks." An extracted object beginning at Week 6, repeating every six weeks to study end, using a ± 3 day window, and anchoring to drug cycles contains four distinct errors. The start and end contradict the text; the Q12W phase is missing; the window was invented; and the anchor requires verification. Structural JSON validation would detect none of these, whereas rule expansion and evidence checks expose them directly.

0.7.3 Conditional events and participant-specific schedules

For a conditional activity a with trigger $C_a(H_t)$ based on history up to time t, planned requiredness is

$$R_a(t) = \mathbb{1}\{C_a(H_t) = 1\}.$$

The source phrase "ECG if clinically indicated" should not be converted into an unconditional ECG or discarded as noncomputable. The record retains the activity, conditional status, trigger language, evidence, and decision owner. A clinician may determine the trigger for an individual participant. If the available record does not establish the trigger, requiredness remains unknown.

Participant-specific schedule generation can be viewed as a state machine. At each transition, treatment status, disease progression, adverse events, withdrawal, or other protocol conditions activate or deactivate future rules. Unscheduled assessments form separate events linked to their clinical reason; they should not be forced onto the nearest planned visit without an explicit analysis convention.

0.7.4 AI-assisted SOA digitization

A schedule agent should distinguish document reconstruction from clinical interpretation:

1. **Locate:** identify the authoritative protocol version, schedule pages, continued tables, legends, and referenced sections.
2. **Recover layout:** reconstruct rows, nested column headers, merged cells, symbols, and footnote identifiers.
3. **Retrieve context:** resolve footnotes and prose references with locations and version scope.
4. **Interpret:** propose activities, timing, conditions, recurrence, stopping rules, exceptions, and uncertainty states.
5. **Normalize and validate:** create stable records, expand recurrence with code, compare every field with evidence, and queue only unresolved cases for review.

Narrow tools make this workflow auditable. A layout parser returns cells and header paths; a retriever returns referenced text; the language model proposes event rules; a schedule engine performs date arithmetic and recurrence expansion; and validators return failures and a review queue. Missing evidence triggers targeted retrieval and revision. If ambiguity remains after the loop stops, it remains visible rather than being converted into a guessed value.

Validation again has distinct layers:

- **structural:** valid identifiers, types, header paths, and resolved references;
- **temporal:** plausible anchors, units, windows, recurrence boundaries, transition points, and stopping conditions;
- **evidential:** every field traces to a cell, header, note, or source sentence in the correct document version;
- **clinical:** conditionality, conflicting instructions, and consequential ambiguity receive expert review.

0.7.5 Evaluation and downstream traceability

Cell-level accuracy is insufficient because not all errors have equal impact. Evaluation should separately score row activities, timepoints, marked cells, header paths, and footnote links; then assess anchor, window, condition, recurrence, and stopping-rule meaning. Schedule behavior can be tested by expanding rules on synthetic anchor dates and comparing the resulting events and participant-specific branches with a reference schedule. Evidence coverage measures the proportion of output fields supported by correctly located source text.

The resulting verified schedule can support visit planning, eCRF design, protocol-deviation review, protocol comparison, and endpoint traceability. For deviation review, the schedule supplies expected events and windows while actual assessment records supply completion and timing. For endpoint work, the lineage should connect

$$\begin{aligned}\text{protocol activity} &\longrightarrow \text{collection field} \longrightarrow \text{source value}\\ &\longrightarrow \text{derived variable} \longrightarrow \text{endpoint analysis.}\end{aligned}$$

This makes it possible to determine whether an absent endpoint component was never planned, conditionally not required, missed operationally, or lost during data processing.

Standards play complementary roles: protocol specifications promote consistent source structure, digital study models represent activities and timelines, and healthcare-exchange standards transport definitions and observations. A definitional plan must still remain distinct from a participant's actual clinical record, and conversion into a standard does not eliminate the need to verify the extracted meaning.

0.8 From Schedules to Time-to-Event Endpoints

The SOA provides opportunities to observe a participant, but it does not itself define an endpoint. Endpoint construction joins five layers:

$$\text{plan} \longrightarrow \text{observation} \longrightarrow \text{clinical assessment} \longrightarrow \text{endpoint derivation} \longrightarrow \text{analysis}.$$

For progression-free survival (PFS), scheduled imaging becomes an acquired scan; measurements and clinical rules produce a response assessment; that assessment is combined with vital status and censoring rules to produce a participant-level time and event indicator. Each arrow requires both data and an authoritative rule.

0.8.1 Defining and deriving progression-free survival

Let R_i be participant i's randomization date, P_i the earliest qualifying objective-progression date under the designated assessment stream, and D_i the qualifying death date. The endpoint event date is

$$E_i = \min(P_i, D_i),$$

where a nonexistent or nonqualifying candidate is treated as $+\infty$. If E_i occurs before the analysis cutoff and satisfies all prespecified rules,

$$\Delta_i = 1, \qquad T_i = E_i - R_i,$$

and the event type records whether $E_i = P_i$ or $E_i = D_i$. If no event qualifies, a prespecified censoring rule produces C_i and

$$\Delta_i = 0, \qquad T_i = C_i - R_i.$$

Whether elapsed time includes the origin day is a study convention and must be implemented consistently. More importantly, C_i cannot be improvised from the last row in a dataset: the SAP determines which adequate disease assessment or follow-up date supports censoring and how missed visits, new anticancer therapy, and incomplete assessments are handled.

The inputs commonly arrive from different systems:

Endpoint component	Typical evidence and role
origin	Randomization system supplies the observed origin date.
progression candidate	Imaging acquisition plus the designated central or investigator assessment supplies date, method, and result.
death candidate	Site, registry, or survival follow-up supplies the vital-status date.
event type and date	Deterministic derivation applies event hierarchy and temporal rules.
censoring date	Derivation combines last adequate status with the approved censoring convention.

Acquisition, review, assigned-result, and data-transfer dates around the same scan serve different purposes. A progression assessment normally uses the clinical assessment date specified by the endpoint definition, not the date on which a vendor transmitted the record.

0.8.1.1 A deterministic derivation algorithm

For each participant, an auditable implementation performs the following:

1. retrieve and validate the endpoint origin;
2. select the governing assessment stream and order all adequate assessments by clinical date;
3. find the earliest qualifying progression candidate;
4. retrieve death and verify its date relative to the origin, cutoff, and other endpoint rules;
5. apply missed-assessment, subsequent-therapy, adequacy, and other qualifiers to both candidates;
6. select the first qualifying event, or apply the prespecified censoring rule if none exists;
7. emit time, indicator, event type, evidence references, rule version, and unresolved review flags.

For example, if a participant is randomized on January 1, has stable disease at Week 12, partial response at Week 18, and centrally confirmed progression on June 19, the PFS event is progression on June 19. If instead the participant has no progression and dies May 25, death is the PFS event. A participant alive at cutoff with no qualifying event is censored according to the approved rule, commonly at a last adequate nonprogressive assessment rather than at the cutoff itself.

0.8.2 RECIST assessments are longitudinal

Tumor imaging becomes an endpoint assessment only after applying a response system such as RECIST 1.1. Its inputs include target-lesion measurements, non-target disease status, and new lesions; the output may be complete response (CR), partial response (PR), stable disease (SD), or progressive disease (PD).

Target-lesion progression depends on a trajectory, not on a single scan. Let S_{ij} be the sum of target-lesion diameters for participant i at assessment j, and let

$$N_{ij} = \min_{k \le j} S_{ik}$$

be the on-study nadir through that assessment. Under the standard target- lesion RECIST rule, PD requires both a relative increase of at least 20% from the nadir and an absolute increase of at least 5 mm:

$$\frac{S_{ij} - N_{ij}}{N_{ij}} \ge 0.20 \quad \text{and} \quad S_{ij} - N_{ij} \ge 5 \text{ mm}.$$

For a nadir of 40 mm and current sum of 50 mm, the increases are 25% and 10 mm, so both conditions hold. Unequivocal progression of non-target disease or a qualifying new lesion can provide other routes to PD. The precise endpoint must state whether investigator assessment or blinded independent central review governs; an available result from the wrong stream is evidence, but not automatically the primary event.

An unscheduled scan may still qualify. “Unscheduled” describes its relation to the visit plan, not its clinical validity. The derivation should preserve why it occurred, its acquisition quality, assessment stream, and applicable protocol rule.

0.8.3 Traceability from activity to endpoint

A PFS record should link each derived field to both its value evidence and the rule that transformed it. A useful traceability graph has the endpoint record as its root, with branches to

- the randomization record supporting origin;
- every progression candidate, its actual scan, assessment result, reviewer stream, and planned or unscheduled context;
- the death candidate and vital-status evidence;
- the governing endpoint, adequacy, event-selection, and censoring rules with document versions.

The same imaging stream may feed PFS, objective response, duration of response, and change in tumor size, but each endpoint uses a different rule. Reuse the evidence, not the endpoint conclusion.

The SOA supplies planned timing, windows, modalities, and follow-up opportunities. It usually does not supply actual dates and results, full progression criteria, the governing reviewer, event hierarchy, or missing-data and censoring rules. An agent must retrieve these semantics from the endpoint section, imaging charter, response criteria, discontinuation rules, eCRF and data specifications, and SAP. It must not manufacture them to complete a schema.

Example 0.8.1 (Evidence without a finalizable primary event)**.** A participant is randomized January 1, has stable disease at Week 12, misses Week 18, and has investigator-assessed PD on an unscheduled May 20 scan. The primary endpoint requires central review, but no central result is available; the participant is known alive through June 30. The May 20 scan and investigator result should be retained as a progression candidate. The primary progression date and any censoring date remain unresolved until the central result is retrieved or the approved endpoint rule specifies how to handle its absence. Schema completeness is not a reason to choose a date.

0.8.4 Assessment time is an observation mechanism

Biological progression generally occurs between assessments. If the last nonprogressive assessment is at L_i and the first progressive assessment at U_i, then the latent progression time P_i^* is interval censored:

$$L_i < P_i^* \leq U_i.$$

The conventional observed date $P_i = U_i$ is a detection time, not necessarily the biological onset. If assessments occur every q time units and latent progression is approximately uniform between visits, the detection delay $W_i = U_i - P_i^*$ satisfies

$$W_i \sim \text{Uniform}(0, q), \qquad \mathbb{E}(W_i) = \frac{q}{2}, \qquad \text{Var}(W_i) = \frac{q^2}{12}.$$

This simple calculation explains why changing from six-week to twelve-week imaging can add roughly three weeks of mean detection delay under idealized conditions.

Assessment frequency can therefore affect reported PFS distributions, event yield by cutoff, and power. If randomized arms have different schedules, or if visit attendance differs by treatment or prognosis, the observation process can distort comparisons. Missing assessments widen the interval containing latent progression and may interact with subsequent therapy or censoring rules. Central review can standardize interpretation of available images, but it cannot recover missing scans, repair poor acquisition, equalize schedules, or restore lost vital-status follow-up.

Treatment discontinuation and endpoint follow-up must also be separated. A participant may stop study treatment yet continue imaging and survival follow-up. Ending all observation at treatment discontinuation loses endpoint information and may induce informative censoring when discontinuation is related to prognosis.

0.8.5 Endpoint–schedule consistency checks

Before a schedule is operationalized, a traceability agent should verify:

1. **coverage:** every endpoint component has a collection path;
2. **duration:** follow-up continues beyond treatment when the endpoint requires it;
3. **reviewer:** the governing assessment stream is explicit;
4. **timing:** nominal, acquisition, assessment, and transfer dates remain distinct;
5. **rules:** events, adequacy, and censoring trace to authoritative versioned text;
6. **balance:** randomized groups have comparable observation opportunities unless a justified analysis addresses the difference.

Common failures include no post-treatment imaging, disagreement between the endpoint section and SOA, an unspecified reviewer stream, no vital-status follow-up, and invented visit windows. A syntactically valid schedule can still encode an invalid or non-identifiable endpoint.

0.8.6 Endpoint-aware schedule optimization

Lean design does not mean minimizing the number of rows or visits at any cost. Endpoint and safety activities are protected; unlinked or duplicative activities are challenged; timing or collection mode may be redesigned; and candidate schedules are evaluated for information, participant burden, feasibility, and auditability.

Let s denote a candidate schedule. One multi-objective formulation is

$$\min_{s \in \mathcal{S}} \left\{ -I(s),\ B_{\text{participant}}(s),\ B_{\text{site}}(s),\ C(s) \right\},$$

where $I(s)$ measures endpoint information, the B terms measure burden, and $C(s)$ measures cost. Feasible schedules must satisfy safety requirements, clinical validity, regulatory constraints, arm balance, and endpoint identifiability. Equivalently, stakeholders may choose weights and minimize

$$J(s) = -w_I I(s) + w_P B_{\text{participant}}(s) + w_S B_{\text{site}}(s) + w_C C(s),$$

but a single weighted score can hide tradeoffs. A Pareto frontier exposes schedules for which no objective can improve without worsening another.

Simulation evaluates candidates before adoption:

1. generate latent progression, death, dropout, treatment stop, and possibly subsequent therapy under clinically plausible scenarios;
2. apply the candidate planned schedule, visit windows, unscheduled assessments, missedness, and review process;
3. derive observed endpoint events and censoring exactly as the future analysis would;
4. compare detection delay, event yield, bias, precision or power, visits, scans, participant time, site workload, and cost;

5. stress-test deviations, differential missingness, and model misspecification before expert and regulatory review.

A schedule with fewer scans may be attractive operationally but unacceptable if it creates long detection delays, insufficient events, differential follow-up, or an endpoint whose derivation cannot be audited. Optimization begins from endpoint requirements, not from row counts.